

Asymptotically Optimal Mixed-State Cloning

Abstract

We determine the asymptotic global fidelity for cloning full-rank nondegenerate finite-dimensional mixed states in the proportional regime $m_n/n \rightarrow \gamma > 1$. For the unitary orbit of a known spectrum $p_1 > \dots > p_d > 0$, the optimal value is

$$F_{\text{orb}}(\gamma, p) = \prod_{i < j} \frac{(1 - q_{ij})(1 - q_{ij}^{(\gamma)})}{(1 - \sqrt{q_{ij}q_{ij}^{(\gamma)}})^2}, \quad q_{ij} = p_j/p_i, \quad q_{ij}^{(\gamma)} = 1 - \frac{1 - q_{ij}}{\gamma}.$$

It is achieved by a Schur–Weyl protocol that measures the Young diagram and applies a Cartan/Parthasarathy–Ranga Rao–Varadarajan (PRV) sector map between typical Weyl irreducible representations. The converse follows from finite-dimensional local asymptotic normality and optimal displaced-thermal Gaussian amplification. When the spectrum is unknown, the same protocol becomes adaptive by choosing the output Young diagram as a rounded dilation of the measured input diagram; the optimal local asymptotic fidelity is

$$F_{\text{univ}}(\gamma, p) = \left(\frac{2\sqrt{\gamma}}{1 + \gamma} \right)^{d-1} F_{\text{orb}}(\gamma, p),$$

where the prefactor is the optimal classical Gaussian amplification fidelity for the spectral LAN component. Thus, in the full-rank nondegenerate regime, the finite-dimensional mixed-state cloning problem admits a sharp asymptotic solution for global all-site fidelity.

1 Introduction

The no-cloning theorem says that an arbitrary unknown quantum state cannot be copied perfectly [?, ?]. Approximate cloning asks instead for the best physical channel which maps n identical input systems into m output systems whose joint state is as close as possible to m ideal copies. For pure states, this problem is well understood: after the first universal qubit machines, Werner gave a covariant construction that solves universal pure-state cloning in arbitrary finite dimension with global fidelity equal to a symmetric-subspace dimension ratio [?, ?, ?, ?]. We recall this construction and its global and one-site fidelities in ???. In the decades since Werner’s theorem, no comparably general solution has been available for genuinely mixed finite-dimensional states with a global all-site product fidelity. The main contribution of this paper is to give such a solution in the asymptotic proportional-output regime, for full-rank nondegenerate states. [Possible referee-sensitive wording: “to our knowledge, the first sharp asymptotic solution for global all-site mixed-state cloning in arbitrary fixed finite dimension.”]

It is important to separate *cloning* from *broadcasting*. A cloning channel aims to approximate the full product state $\rho^{\otimes m}$. A broadcasting channel only asks that selected one-site marginals equal, or approximate, the input state; the joint output may be strongly correlated and far from a product. The single-copy no-broadcasting theorem says that a set of states can be broadcast exactly only when

the states commute [?]. This theorem should not be confused with multi-input broadcasting: with several input copies, the obstruction is different. In particular, D’Ariano–Macchiavello–Perinotti showed for mixed qubit states that the no-broadcasting theorem does not extend directly beyond a single input copy, and that for four or more input copies one can even increase the local purity while broadcasting, a phenomenon they called superbroadcasting [?]. Subsequent work developed universal and phase-covariant superbroadcasting and optimal mixed-qubit broadcasting maps [?, ?, ?]. In regimes where the one-site output is more pure than the input, exact marginal broadcasting is obtained by adding local depolarizing noise; this still does not produce independent product copies. For displaced thermal oscillator states, D’Ariano–Perinotti–Sacchi also give an explicit exact broadcasting construction when the input thermal photon number is known [?]. These results are about marginal outputs, not global product fidelity. They therefore do not solve the cloning problem studied here.

Much of the previous finite-dimensional approximate mixed-state cloning and broadcasting literature focuses on qubits, special covariance classes, purification, partial cloning, or marginal figures of merit. Cirac–Ekert–Macchiavello introduced a decomposition of $\rho^{\otimes N}$ for optimal single-qubit purification and noted its connection with optimal cloning and estimation of mixed states [?]. Rastegin developed global-fidelity, relative-error, and state-dependent bounds for mixed-state cloning [?, ?, ?]; Fan constructed cloners for mixed states supported on symmetric subspaces and for identical mixed qubits [?, ?]; and Kazakov studied partial cloning and mixed-state cloning in a related finite-dimensional setting [?]. The LAN approach has also been used for asymptotic purification and dilution of mixed qubits [?]. Continuous-variable cloning provides another comparison point: optimal coherent-state cloners are realized by linear amplification and beam splitters [?, ?], and Guta–Matsumoto solved optimal cloning of displaced mixed Gaussian states by reducing it to least-noise amplification [?]. Our converse uses precisely this Gaussian amplification theorem after finite-dimensional local asymptotic normality reduces the mixed-state orbit to independent oscillator modes.

We consider two finite-dimensional regimes. In the known-spectrum problem, the spectrum $p = (p_1, \dots, p_d)$, $p_1 > \dots > p_d > 0$, is fixed and known, and one clones the unitary orbit

$$\rho_{p,g} = U(g) \text{diag}(p) U(g)^*.$$

Only the eigenbasis parameter is unknown. In the unknown-spectrum problem, the spectrum also varies locally; the Schur–Weyl Young label then carries a classical Gaussian LAN component. The known-spectrum model contains only the off-diagonal quantum oscillator modes, whereas the unknown-spectrum model contains both the classical spectral Gaussian part and the quantum orbit part.

The abbreviation PRV refers to the theorem of Parthasarathy, Ranga Rao, and Varadarajan on extremal components in tensor products of irreducible representations. We use its Cartan-component special case: if α and β are dominant weights, then $\mathcal{H}_{\alpha+\beta}$ appears canonically in $\mathcal{H}_\alpha \otimes \mathcal{H}_\beta$. The associated normalized inclusion is the PRV inclusion. Our sector channel is the adjoint, trace-normalized operation obtained by embedding an output irrep $\mathcal{H}_\nu$ into $\mathcal{H}_\mu \otimes \mathcal{H}_{\nu-\mu}$ and tracing against the auxiliary factor. A formal definition is given in ?? before the protocol is stated.

We use the global, all-site fidelity

$$F(\sigma, \tau) = \|\sqrt{\sigma}\sqrt{\tau}\|_1^2$$

between the actual m -partite output and the ideal state $\rho^{\otimes m}$. This is stricter than one-site marginal fidelity and is the key distinction from the broadcasting/superbroadcasting literature. We work in the proportional-output regime

$$\frac{m_n}{n} \longrightarrow \gamma > 1.$$

1.1 Informal statement of the main results

Let

$$\rho_p = \text{diag}(p_1, \dots, p_d), \quad p_1 > \dots > p_d > 0,$$

and let $\rho_{p,g} = U(g)\rho_p U(g)^*$. For every positive root $i < j$, set

$$q_{ij} = \frac{p_j}{p_i}, \quad q_{ij}^{(\gamma)} = 1 - \frac{1 - q_{ij}}{\gamma}.$$

The known-spectrum answer is

$$F_{\text{orb}}(\gamma, p) = \prod_{i < j} \frac{(1 - q_{ij})(1 - q_{ij}^{(\gamma)})}{(1 - \sqrt{q_{ij}q_{ij}^{(\gamma)}})^2}. \quad (1)$$

Equivalently,

$$F_{\text{orb}}(\gamma, p) = \prod_{i < j} \left(\frac{\sqrt{\gamma} + \sqrt{q_{ij}(\gamma - 1 + q_{ij})}}{\gamma + q_{ij}} \right)^2. \quad (2)$$

The Schur–PRV protocol achieves (??). Conversely, any global channel that beat (??) would, through the Kahn–Guta LAN maps, induce a displaced-thermal Gaussian cloner beating the least-noise Gaussian amplifier. That contradicts the Gaussian cloning theorem of Guta–Matsumoto together with the fidelity optimality argument given below [?, ?].

If the spectrum is unknown, the optimal local asymptotic fidelity at a fixed nondegenerate spectrum p becomes

$$F_{\text{univ}}(\gamma, p) = \left(\frac{2\sqrt{\gamma}}{1 + \gamma} \right)^{d-1} F_{\text{orb}}(\gamma, p). \quad (3)$$

The extra factor is purely classical. It is the squared Hellinger fidelity between the output Young-label Gaussian produced by optimally amplifying the measured Schur label and the target Young-label Gaussian. Thus measuring the Young diagram and adapting the PRV sector map is optimal, but it cannot remove the $n^{-1/2}$ -scale uncertainty in the spectrum unless the spectrum is given as side information.

1.2 Related work and comparison

The closest existing finite-dimensional all-site construction is the random-purification plus pure-state-cloning protocol of Li–Theil–Harrow–Chuang [?]. It is universal over rank- r mixed states: one first purifies the input by a random-purification channel, applies Werner’s optimal pure-state cloner in dimension $D = dr$, and traces out the purifying registers. Thus its stated all-site fidelity guarantee is precisely the Werner global-fidelity dimension ratio (??) with $D = dr$:

$$F_{\text{RP}}(n, m; d, r) \geq \frac{\binom{n+dr-1}{dr-1}}{\binom{m+dr-1}{dr-1}}, \quad (4)$$

so, for $m/n \rightarrow \gamma$,

$$F_{\text{RP}}^\infty(\gamma; d, r) \geq \gamma^{-(dr-1)}.$$

For full rank $r = d$, this is $\gamma^{-(d^2-1)}$. In contrast, our fixed-spectrum orbit fidelity obeys

$$F_{\text{orb}}(\gamma, p) > \gamma^{-d(d-1)/2},$$

with strict inequality for every $\gamma > 1$ and full-rank p . Even in the unknown-spectrum problem,

$$F_{\text{univ}}(\gamma, p) > \left(\frac{2\sqrt{\gamma}}{1+\gamma} \right)^{d-1} \gamma^{-d(d-1)/2} > \gamma^{-(d^2-1)}.$$

This comparison should be read carefully: (??) is a universal rank- r guarantee for a different, harder task, not the exact fidelity of that channel on a fixed-spectrum orbit.

Other mixed-state cloning and broadcasting results address different operational criteria. Rastegin’s bounds apply to mixed-state cloning, relative-error criteria, and state-dependent cloning, but they do not identify the sharp arbitrary-dimensional proportional-output all-site asymptotics considered here [?, ?, ?]. Fan’s mixed-state cloners treat symmetric-subspace inputs and identical mixed qubits, again with criteria different from our global product fidelity [?, ?]; Kazakov’s partial-cloning construction is another qubit-level mixed-state copying model [?]. The superbroadcasting literature optimizes one-site marginals of correlated outputs; it demonstrates that marginal broadcasting can be exact, or even locally purifying, when multiple input copies are available, but it does not require the joint output to approximate $\rho^{\otimes m}$ [?, ?, ?, ?]. Mixed Gaussian cloning, by contrast, is very close in spirit to our LAN limit; Guta–Matsumoto’s displaced-thermal amplifier theorem is the Gaussian converse used in this paper [?].

[Possible addition for final introduction: mention explicitly that the finite- n universal mixed-state analogue of Werner’s theorem remains open; our statement is the sharp asymptotic proportional-output solution for the full-rank nondegenerate finite-dimensional regime.]

1.3 Organization

The paper is organized as follows. ?? fixes notation, recalls Werner’s pure-state cloner, reviews the Schur–Weyl block form, Young concentration, PRV sector maps, LAN models, and the displaced-thermal Gaussian fidelity converse. ?? gives the known-spectrum Schur–PRV protocol, its achievability proof, and the global converse. ?? treats the adaptive unknown-spectrum protocol, the expanding-window local converse, and the compact strict-chamber global corollary. ?? compares the resulting fidelities with random purification followed by pure-state cloning, and ?? discusses extensions.

[Internal revision note: earlier drafts used a stronger PRV-to-amplifier channel-convergence assumption and kept several proof-status appendices. In this version the known-spectrum achievability uses only the thermal PRV sector theorem, and the unknown-spectrum converse uses the hybrid product converse proved in the appendix. Finite- n rates and boundary/degenerate spectra remain outside the stated theorem.]

2 Preliminaries and Gaussian amplification

2.1 Fidelity conventions

We use squared Uhlmann fidelity

$$F(\rho, \sigma) = \|\sqrt{\rho}\sqrt{\sigma}\|_1^2$$

and root fidelity

$$B(\rho, \sigma) = \sqrt{F(\rho, \sigma)}.$$

For classical probability distributions P, Q , root fidelity is the Hellinger affinity

$$B(P, Q) = \sum_x \sqrt{P(x)Q(x)}$$

or the corresponding integral. The squared classical fidelity is $F(P, Q) = B(P, Q)^2$.

We will use the following standard facts.

Lemma 2.1 (Continuity and monotonicity). *Let Λ be a quantum channel. Then*

$$B(\Lambda(\rho), \Lambda(\sigma)) \geq B(\rho, \sigma).$$

Root fidelity is jointly concave. Moreover, if $\|\rho - \rho'\|_1 \leq \varepsilon$ and $\|\sigma - \sigma'\|_1 \leq \varepsilon$, then

$$|B(\rho, \sigma) - B(\rho', \sigma')| \leq C\sqrt{\varepsilon}$$

for a universal constant C .

Proof. Monotonicity and joint concavity are standard consequences of Uhlmann's theorem. The continuity bound follows from the Fuchs–van de Graaf inequalities and the triangle inequality for the Bures distance

$$d_B(\rho, \sigma) = \sqrt{2(1 - B(\rho, \sigma))}.$$

Indeed $d_B(\rho, \rho') \leq \|\rho - \rho'\|_1^{1/2}$, and similarly for σ, σ' . Since $B = 1 - d_B^2/2$, this gives the displayed estimate for a universal constant; only convergence to zero is used in the LAN transfers below. $\square$

If states are block diagonal over a common classical label,

$$\rho = \bigoplus_x P_x \rho_x, \quad \sigma = \bigoplus_x Q_x \sigma_x,$$

then

$$B(\rho, \sigma) = \sum_x \sqrt{P_x Q_x} B(\rho_x, \sigma_x). \quad (5)$$

2.2 Werner's universal pure-state cloner

We recall the pure-state benchmark that underlies much of the comparison with mixed-state constructions. Let the single-system Hilbert space have dimension D , and let

$$P_{\text{sym}}^{(r)} : (\mathbb{C}^D)^{\otimes r} \rightarrow \text{Sym}^r(\mathbb{C}^D)$$

be the projector onto the totally symmetric subspace. Write

$$d_{r,D}^{\text{sym}} := \dim \text{Sym}^r(\mathbb{C}^D) = \binom{r + D - 1}{D - 1}. \quad (6)$$

Werner's optimal universal pure-state $n \rightarrow m$ cloner is the covariant channel

$$\mathcal{W}_{n \rightarrow m}^{(D)}(X) = \frac{d_{n,D}^{\text{sym}}}{d_{m,D}^{\text{sym}}} P_{\text{sym}}^{(m)}(X \otimes I^{\otimes(m-n)}) P_{\text{sym}}^{(m)}, \quad (7)$$

where X is supported on $\text{Sym}^n(\mathbb{C}^D)$ [?]. The prefactor makes the map trace preserving. For every unknown pure state ψ , its all-site/global fidelity is

$$F_{\text{glob}}^{\text{Werner}}(n, m, D) = \langle \psi |^{\otimes m} \mathcal{W}_{n \rightarrow m}^{(D)}(|\psi\rangle\langle\psi|^{\otimes n}) | \psi \rangle^{\otimes m} = \frac{d_{n,D}^{\text{sym}}}{d_{m,D}^{\text{sym}}} = \frac{\binom{n+D-1}{D-1}}{\binom{m+D-1}{D-1}}. \quad (8)$$

The one-site reduced fidelity of the same universal cloner is

$$F_1^{\text{Werner}}(n, m, D) = \frac{n(m+D) + m - n}{m(n+D)}. \quad (9)$$

For example, $n = 1, m = 2, D = 2$ gives the familiar 5/6 single-qubit fidelity. The distinction between (??) and (??) matters: this paper uses the global/all-site figure of merit, so the relevant pure-state comparator is the dimension ratio (??), not the one-site fidelity. In particular, the random-purification construction discussed in ?? inherits exactly (??) with $D = dr$.

2.3 Schur–Weyl decomposition and Weyl-sector states

Schur–Weyl duality gives

$$(\mathbb{C}^d)^{\otimes n} \cong \bigoplus_{\lambda \vdash n, \ell(\lambda) \leq d} \mathcal{H}_\lambda \otimes \mathcal{S}_\lambda, \quad (10)$$

where $\mathcal{H}_\lambda$ is the irreducible $U(d)$ -module of highest weight λ , and $\mathcal{S}_\lambda$ is the corresponding Specht module. For a diagonal density matrix $\rho_p = \text{diag}(p_1, \dots, p_d)$,

$$\rho_p^{\otimes n} = \bigoplus_{\lambda \vdash n} P_{n,p}(\lambda) \rho_{p,\lambda} \otimes \frac{I_{\mathcal{S}_\lambda}}{\dim \mathcal{S}_\lambda}, \quad (11)$$

where

$$P_{n,p}(\lambda) = \dim(\mathcal{S}_\lambda) s_\lambda(p), \quad \rho_{p,\lambda} = \frac{\pi_\lambda(\rho_p)}{s_\lambda(p)}. \quad (12)$$

Here s_λ is the Schur polynomial and π_λ is the polynomial representation. For an orbit point $\rho_{p,g} = U(g)\rho_p U(g)^*$, the same decomposition holds with

$$\rho_{p,g,\lambda} = U_\lambda(g)\rho_{p,\lambda}U_\lambda(g)^*.$$

The Young distribution $P_{n,p}$ depends on the spectrum p , but not on the eigenbasis g .

Lemma 2.2 (Young concentration). *Fix a compact subset of the nondegenerate full-rank Weyl chamber,*

$$\Delta_{d,\kappa} = \{p : p_d \geq \kappa, p_i - p_{i+1} \geq \kappa, \sum_i p_i = 1\}.$$

There are constants $c, C > 0$, depending only on d and κ , such that for all $p \in \Delta_{d,\kappa}$,

$$P_{n,p} \left(\left\| \frac{\lambda}{n} - p \right\|_1 > \delta \right) \leq C(n+1)^{d(d-1)/2} \exp(-cn\delta^2) \quad (13)$$

for all sufficiently small $\delta > 0$. In particular, for

$$\mathsf{T}_n(p) = \left\{ \lambda : \left\| \frac{\lambda}{n} - p \right\|_\infty \leq n^{-1/3} \right\},$$

one has $P_{n,p}(\mathsf{T}_n(p)^c) \leq \exp(-c'n^{1/3})$.

Proof. This is the Keyl–Werner spectrum-estimation bound applied to the Young-frame measurement [?]. The polynomial prefactor and the exponential rate follow from their large-deviation estimate. The final statement follows by setting $\delta = dn^{-1/3}$ and absorbing the polynomial factor into the exponential. $\square$

2.4 The PRV component and the sector channel

We record the representation-theoretic object used throughout the protocol. The acronym PRV stands for Parthasarathy–Ranga Rao–Varadarajan. The full PRV theorem describes extremal irreducible summands in tensor products of highest-weight representations [?]. The only case needed here is the Cartan, or top, component. If α and β are dominant weights for $U(d)$, then

$$\mathcal{H}_{\alpha+\beta} \subset \mathcal{H}_\alpha \otimes \mathcal{H}_\beta \quad (14)$$

with multiplicity one. We call the corresponding normalized intertwining isometry

$$V_{\alpha,\beta} : \mathcal{H}_{\alpha+\beta} \hookrightarrow \mathcal{H}_\alpha \otimes \mathcal{H}_\beta, \quad V_{\alpha,\beta} v_{\alpha+\beta} = v_\alpha \otimes v_\beta, \quad (15)$$

the PRV inclusion. Thus the phrase “PRV channel” in this paper always refers to the Cartan/PRV component; no other PRV summand is used.

If $\mu \vdash n$, $\nu \vdash m$, and $\omega := \nu - \mu$ is dominant, then $\nu = \mu + \omega$ and (??) gives

$$V_{\mu,\omega} : \mathcal{H}_\nu \hookrightarrow \mathcal{H}_\mu \otimes \mathcal{H}_\omega. \quad (16)$$

The associated sector channel, used later in ??, is

$$C_{\mu \rightarrow \nu}^{\text{PRV}}(X) = \frac{\dim \mathcal{H}_\mu}{\dim \mathcal{H}_\nu} V_{\mu,\omega}^* (X \otimes I_\omega) V_{\mu,\omega}, \quad X \in \mathcal{B}(\mathcal{H}_\mu). \quad (17)$$

This is a channel from $\mathcal{B}(\mathcal{H}_\mu)$ to $\mathcal{B}(\mathcal{H}_\nu)$. Complete positivity is immediate from the Kraus form obtained by inserting an orthonormal basis of $\mathcal{H}_\omega$; trace preservation follows from Schur’s lemma because $\text{Tr}_\omega(V_{\mu,\omega} V_{\mu,\omega}^*)$ is a scalar multiple of the identity on the irreducible space $\mathcal{H}_\mu$, and its trace fixes the scalar to $\dim \mathcal{H}_\nu / \dim \mathcal{H}_\mu$. This early definition is repeated in the protocol section only to keep the construction self-contained.

2.5 Oscillator notation and thermal fidelities

Let $\mathcal{F}$ be one-mode Fock space with annihilation operator a . For $0 < q < 1$, define the thermal state

$$\tau_q = (1 - q) \sum_{k \geq 0} q^k |k\rangle\langle k|. \quad (18)$$

The root fidelity between two such thermal states is classical:

$$B(\tau_q, \tau_{q'}) = \sum_{k \geq 0} \sqrt{(1 - q)(1 - q')q^k(q')^k} = \frac{\sqrt{(1 - q)(1 - q')}}{1 - \sqrt{qq'}}. \quad (19)$$

Thus

$$F(\tau_q, \tau_{q'}) = \frac{(1 - q)(1 - q')}{(1 - \sqrt{qq'})^2}. \quad (20)$$

The quantum-limited phase-insensitive amplifier of gain $G \geq 1$ has the Stinespring form

$$a_{\text{out}} = \sqrt{G} a_{\text{in}} + \sqrt{G-1} b_{\text{env}}^*, \quad (21)$$

with the environment in the vacuum. It sends

$$D(z)\tau_q D(z)^* \mapsto D(\sqrt{G}z)\tau_{q_G} D(\sqrt{G}z)^*, \quad q_G = 1 - \frac{1-q}{G}. \quad (22)$$

For $G = \gamma$, this is exactly $q^{(\gamma)}$.

2.6 LAN models

For a full-rank finite-dimensional state, the Kahn–Guta local asymptotic normality theorem says that the n -copy model converges to a product of a classical Gaussian model for eigenvalue perturbations and independent displaced thermal oscillators for off-diagonal perturbations [?]. We use two specializations.

Theorem 2.3 (Fixed-spectrum LAN). *Fix $p \in \Delta_d^\circ$. Let $R = d(d-1)/2$. There is a local coordinate map $z \in \mathbb{C}^R \mapsto X(z) \in \mathfrak{u}(d)$ for off-diagonal tangent directions such that, with*

$$g_{n,z} = \exp(X(z)/\sqrt{n}), \quad \rho_{n,z} = \rho_{p,g_{n,z}}^{\otimes n},$$

there exist channels T_n and S_n satisfying, uniformly for z in compact subsets of $\mathbb{C}^R$,

$$\|T_n(\rho_{n,z}) - \Phi_z\|_1 \rightarrow 0, \quad (23)$$

$$\|S_n(\Phi_z) - \rho_{n,z}\|_1 \rightarrow 0, \quad (24)$$

where

$$\Phi_z = \bigotimes_{i < j} D(z_{ij})\tau_{q_{ij}} D(z_{ij})^*. \quad (25)$$

For $m_n/n \rightarrow \gamma$, the corresponding output target satisfies

$$T_{m_n}(\rho_{p,g_{n,z}}^{\otimes m_n}) \rightarrow \Psi_z := \bigotimes_{i < j} D(\sqrt{\gamma}z_{ij})\tau_{q_{ij}} D(\sqrt{\gamma}z_{ij})^* \quad (26)$$

in trace norm, uniformly on compact sets. The same group element is written with the input normalization $g_{n,z} = \exp(X(z)/\sqrt{n})$; in the m_n -copy LAN chart this corresponds to the local parameter $\sqrt{m_n/n} z \rightarrow \sqrt{\gamma}z$, which is the source of the displacement gain. The thermal parameter remains q_{ij} , since it depends only on the underlying spectrum p .

Theorem 2.4 (Full mixed-state LAN). *Fix $p \in \Delta_d^\circ$. Let $V_p = \text{diag}(p) - pp^T$, restricted to the tangent space*

$$\mathbf{H} = \{h \in \mathbb{R}^d : \sum_i h_i = 0\}.$$

For local perturbations

$$p_{n,h} = p + h/\sqrt{n}, \quad g_{n,z} = \exp(X(z)/\sqrt{n}),$$

the n -copy model converges, uniformly for (h, z) in compact subsets of $\mathbf{H} \times \mathbb{C}^R$, to

$$\mathbf{N}_{h,V_p} \otimes \Phi_z, \quad (27)$$

where N_{h,V_p} denotes a classical normal distribution on H with mean h and covariance V_p . The m_n -copy target corresponds to

$$N_{\sqrt{\gamma}h,V_p} \otimes \Psi_z. \quad (28)$$

There are forward and inverse LAN channels implementing this convergence in Le Cam deficiency, uniformly on compact sets.

Corollary 2.5 (Compact-uniform LAN on strict chambers). *Fix $\kappa > 0$ and $L < \infty$. The LAN convergence in ??? is uniform for all base spectra*

$$p \in \Delta_{d,\kappa} = \{p : p_d \geq \kappa, p_i - p_{i+1} \geq \kappa, \sum_i p_i = 1\}$$

and all local parameters $(h, z) \in C_L$ for which $p + h/\sqrt{n}$ is a state. In particular, the forward, inverse, and output LAN errors can be chosen so that their suprema over $p \in \Delta_{d,\kappa}$ and $(h, z) \in C_L$ tend to zero.

Proof. This is a compactness consequence of the Kahn–Guta LAN theorem. Their construction is local and uniform on compact parameter sets inside the full-rank state space. On $\Delta_{d,\kappa}$, all eigenvalues and all eigenvalue gaps are bounded away from zero, so the coordinate charts, covariance matrices, oscillator temperatures, and error estimates appearing in the LAN proof vary continuously with uniformly bounded derivatives. Cover $\Delta_{d,\kappa}$ by finitely many such LAN neighborhoods and take the maximum of the corresponding error functions. Since each tends to zero, the finite maximum tends to zero. The fixed-spectrum orbit statement is the restriction of the same uniform full model to the off-diagonal coordinates. $\square$

[Internal citation note: the compactness argument above uses that the Kahn–Guta construction is locally uniform, with LAN channels chosen consistently on each chart; cite the locally uniform statements of [?] explicitly in the final version.]

The fixed-spectrum theorem is obtained from the full theorem by freezing the classical eigenvalue parameter. The unknown-spectrum result below uses the full hybrid model.

2.7 LAN fidelity transfer

We will repeatedly use the following elementary consequence of trace-norm LAN convergence. Suppose finite-dimensional local models $\rho_{n,\theta}$, $\sigma_{n,\theta}$ and Gaussian models Φ_θ , Ψ_θ satisfy, uniformly on a compact parameter set K ,

$$\sup_{\theta \in K} \|S_n(\Phi_\theta) - \rho_{n,\theta}\|_1 \rightarrow 0, \quad \sup_{\theta \in K} \|T_n(\sigma_{n,\theta}) - \Psi_\theta\|_1 \rightarrow 0.$$

For an arbitrary finite-dimensional channel $\mathcal{M}_n$, set

$$\Gamma_n = T_n \circ \mathcal{M}_n \circ S_n.$$

Then

$$\inf_{\theta \in K} B(\Gamma_n(\Phi_\theta), \Psi_\theta) \geq \inf_{\theta \in K} B(\mathcal{M}_n(\rho_{n,\theta}), \sigma_{n,\theta}) - o(1). \quad (29)$$

Indeed, replace $S_n(\Phi_\theta)$ by $\rho_{n,\theta}$ using trace-norm contractivity and ??; apply monotonicity under T_n ; then replace $T_n(\sigma_{n,\theta})$ by Ψ_θ . Thus a finite-dimensional cloner whose fidelity is uniformly above a Gaussian benchmark on a local window induces, through the LAN maps, a Gaussian cloner with the same excess fidelity up to $o(1)$. This is the finite-dimensional-to-Gaussian contradiction template used in the known-spectrum and unknown-spectrum converses below.

2.8 Gaussian amplification and the fidelity converse

This section proves the Gaussian root-fidelity bound used by the finite-dimensional converse. The structural input is the one-mode stochastic-ordering theorem of Guta–Matsumoto for displaced thermal amplification [?]. We use it in a conic form and tensorize it by a peeling argument, so the proof does not require a separate multivariate stochastic-order theorem.

Lemma 2.6 (Uniform trace reduction for covariant CP maps). *Let $\mathcal{J}$ be a normal completely positive trace-nonincreasing map on an oscillator trace class. Suppose $\mathcal{J}$ is displacement covariant with gain $\sqrt{\gamma}$:*

$$\mathcal{J}(D(z)XD(z)^*) = D(\sqrt{\gamma}z)\mathcal{J}(X)D(\sqrt{\gamma}z)^*$$

for all displacements z . Then there is a constant $c \in [0, 1]$ such that

$$\text{Tr } \mathcal{J}(X) = c \text{Tr } X$$

for every trace-class operator X .

Proof. The output trace is a positive normal linear functional on the trace class, so $\text{Tr } \mathcal{J}(X) = \text{Tr}[AX]$ for a bounded operator $0 \leq A \leq I$. Covariance and cyclicity give $D(z)^*AD(z) = A$ for all z . The Weyl representation is irreducible, so its von Neumann commutant is the scalar algebra. Hence $A = cI$, with $c \in [0, 1]$, proving the claim. $\square$

Lemma 2.7 (Hunt–Stein reduction for the Gaussian covariant problems). *For the flat-prior Gaussian cloning problems considered in this paper, with covariance group $\mathbb{C}^R$ in the fixed-spectrum case, $\mathbb{H}$ in the classical spectral case, and $\mathbb{H} \times \mathbb{C}^R$ in the hybrid unknown-spectrum case, the expanding-window minimax value is unchanged if the supremum over all channels is restricted to exactly covariant channels.*

Proof. The covariance groups are locally compact abelian, hence amenable. The payoff is root fidelity averaged over a Haar-uniform prior on expanding Følner cubes, and root fidelity is concave in the channel output. The Hunt–Stein theorem for amenable covariance groups therefore gives an invariant, equivalently covariant, decision rule with value at least the limiting flat-prior value of any noncovariant rule; see the standard quantum decision-theoretic formulation in Holevo’s treatment of covariant statistical decision problems [?]. Equivalently, one may average a channel over Følner cubes and use joint concavity; the amenability theorem is the clean way to pass from approximate covariance of the averages to an exactly covariant optimizer without weak-limit normalization subtleties. $\square$

2.8.1 One oscillator mode

Fix $0 < q < 1$ and $\gamma > 1$. The input family is

$$\Phi_z = D(z)\tau_q D(z)^*,$$

and the ideal cloning target is

$$\Psi_z = D(\sqrt{\gamma}z)\tau_q D(\sqrt{\gamma}z)^*.$$

The least-noise amplifier produces

$$D(\sqrt{\gamma}z)\tau_{q^{(\gamma)}} D(\sqrt{\gamma}z)^*, \quad q^{(\gamma)} = 1 - \frac{1-q}{\gamma}.$$

Its squared fidelity with the target is (??) with $q' = q^{(\gamma)}$.

Lemma 2.8 (One-mode conic stochastic order). *Let $\mathcal{J}$ be a normal completely positive trace-nonincreasing map on one oscillator mode, displacement covariant with gain $\sqrt{\gamma}$, and phase covariant. Then, for every bounded nonnegative decreasing function $w : \mathbb{N} \rightarrow \mathbb{R}_+$,*

$$\text{Tr}[\mathcal{J}(\tau_q)w(N)] \leq \text{Tr}[\mathcal{J}(\tau_q)] \text{Tr}[\tau_{q^{(\gamma)}}w(N)]. \quad (30)$$

Proof. The trace functional of $\mathcal{J}$ has the form $X \mapsto \text{Tr}[AX]$, with $0 \leq A \leq I$. Exact displacement covariance and cyclicity of trace imply $D(z)^*AD(z) = A$ for all Weyl displacements. By irreducibility of the Weyl representation, $A = cI$, so $\text{Tr} \mathcal{J}(X) = c \text{Tr} X$. If $c = 0$, the claim is trivial. If $c > 0$, then $c^{-1}\mathcal{J}$ is a trace-preserving phase-insensitive displacement-covariant amplifier of gain γ . The Guta–Matsumoto stochastic-ordering theorem says that its output photon-number distribution on τ_q is stochastically larger than the least-noise geometric distribution with parameter $q^{(\gamma)}$. Therefore the expectation of every bounded decreasing photon-number function is no larger than for $\tau_{q^{(\gamma)}}$. Multiplying by c gives (??). $\square$

Lemma 2.9 (Fidelity optimality of the least-noise amplifier). *Among displacement-covariant amplifiers of gain γ , the root fidelity with the target displaced thermal family is maximized by the quantum-limited amplifier. The optimal squared fidelity is*

$$f_\gamma(q) = \frac{(1-q)(1-q^{(\gamma)})}{(1-\sqrt{q}q^{(\gamma)})^2}. \quad (31)$$

Proof. By displacement covariance, it suffices to evaluate the output at $z = 0$. Since τ_q is phase invariant, phase-twirling an output cannot decrease root fidelity, by joint concavity and phase invariance. Hence the optimizer may be assumed diagonal, say $\sigma = \sum_k P_k |k\rangle\langle k|$. Let $P_k^* = (1-q^{(\gamma)})(q^{(\gamma)})^k$ and $T_k = (1-q)q^k$. The root fidelity is $B(P, T) = \sum_k \sqrt{P_k T_k}$. At P^* , the first variation has weights

$$w_k = \sqrt{T_k/P_k^*} = \sqrt{\frac{1-q}{1-q^{(\gamma)}}} \left(\frac{q}{q^{(\gamma)}} \right)^{k/2},$$

which are decreasing. Applying ?? to this bounded decreasing weight gives

$$\sum_k w_k (P_k - P_k^*) \leq 0.$$

Concavity of $P \mapsto B(P, T)$ then gives

$$B(P, T) \leq B(P^*, T) + \frac{1}{2} \sum_k w_k (P_k - P_k^*) \leq B(P^*, T).$$

The value $B(P^*, T)$ is (??) with $q' = q^{(\gamma)}$, proving (??). $\square$

The equivalent closed form in (??) follows directly from $q^{(\gamma)} = (\gamma - 1 + q)/\gamma$. Since $1 - q^{(\gamma)} = (1 - q)/\gamma$,

$$f_\gamma(q) = \frac{(1-q)^2/\gamma}{(1-\sqrt{q}(\gamma-1+q)/\gamma)^2} = \left(\frac{\sqrt{\gamma} + \sqrt{q(\gamma-1+q)}}{\gamma + q} \right)^2.$$

Multiplying this identity over all positive roots gives (??).

2.8.2 Tensorization and flat-prior minimax form

For the fixed-spectrum orbit, the Gaussian input and target are Φ_z and Ψ_z from (??) and (??). Define

$$B_G(\gamma, p) := \prod_{i < j} \frac{\sqrt{(1 - q_{ij})(1 - q_{ij}^{(\gamma)})}}{1 - \sqrt{q_{ij}q_{ij}^{(\gamma)}}}, \quad (32)$$

$$F_G(\gamma, p) := B_G(\gamma, p)^2. \quad (33)$$

This is the same quantity as $B_{\text{orb}}, F_{\text{orb}}$.

Lemma 2.10 (Tensorization of the one-mode conic order). *Let $\mathcal{A}$ be an R -mode normal completely positive trace-nonincreasing map, displacement covariant with gain $\sqrt{\gamma}$ in every mode and phase covariant in every mode. Let*

$$\Phi_0 = \bigotimes_{\alpha=1}^R \tau_{q_\alpha}, \quad \Omega_{*,Q} = \bigotimes_{\alpha=1}^R \tau_{q_\alpha^{(\gamma)}}.$$

For every bounded product weight $W(\mathbf{N}) = \prod_{\alpha=1}^R w_\alpha(N_\alpha)$, with each $w_\alpha \geq 0$ decreasing,

$$\text{Tr}[\mathcal{A}(\Phi_0)W(\mathbf{N})] \leq \text{Tr}[\mathcal{A}(\Phi_0)] \text{Tr}[\Omega_{*,Q}W(\mathbf{N})]. \quad (34)$$

In particular, if $\mathcal{A}$ is trace preserving, then the prefactor $\text{Tr}[\mathcal{A}(\Phi_0)]$ is one.

Proof. By homogeneity, scale the weights so that $0 \leq w_\alpha \leq 1$. Write $W_{>1} = \prod_{\alpha=2}^R w_\alpha(N_\alpha)$. From $\mathcal{A}$, construct a one-mode completely positive trace-nonincreasing map $\mathcal{J}_1$ by the dual identity

$$\text{Tr}[\mathcal{J}_1(X)Y] := \text{Tr}[\mathcal{A}(X \otimes \Phi_{0,>1})(Y \otimes W_{>1})], \quad 0 \leq Y \in \mathcal{B}(\mathcal{F}_1).$$

Since $0 \leq W_{>1} \leq I$ and $\mathcal{A}$ is trace-nonincreasing, $\mathcal{J}_1$ is trace-nonincreasing. It is displacement covariant with gain $\sqrt{\gamma}$ and phase covariant in the first mode, because $\mathcal{A}$ has those covariances and $W_{>1}$ commutes with the phase rotations of the other output modes. Applying ?? gives

$$\text{Tr}[\mathcal{A}(\Phi_0)w_1(N_1)W_{>1}] \leq \text{Tr}[\tau_{q_1^{(\gamma)}}w_1(N_1)] \text{Tr}[\mathcal{A}(\Phi_0)W_{>1}].$$

Repeating the same argument for modes $2, \dots, R$ gives

$$\text{Tr}[\mathcal{A}(\Phi_0)W] \leq \left(\prod_{\alpha=1}^R \text{Tr}[\tau_{q_\alpha^{(\gamma)}}w_\alpha(N_\alpha)] \right) \text{Tr}[\mathcal{A}(\Phi_0)],$$

which is exactly (??). □

Remark 2.11 (The single external analytic input). *After the tensorization above, the entire Gaussian converse rests on a single one-mode statement, namely ??: for a trace-preserving, phase-insensitive, displacement-covariant map of gain γ , the output photon-number distribution on τ_q stochastically dominates the least-noise geometric law with parameter $q^{(\gamma)}$. We import this stochastic-ordering statement from the displaced-thermal amplification analysis of Guta–Matsumoto [?]. The logical order matters: the fidelity optimality in ?? is derived from the stochastic order, not conversely. No separate multivariate stochastic-order theorem is needed, since the multimode case is reduced to the one-mode statement by the peeling argument of ??.*

[Internal citation note: pin the precise statement in Guta–Matsumoto that asserts the photon-number stochastic ordering for covariant gain- γ channels, as opposed to its fidelity-optimality corollary. If their published form is fidelity optimality only, the one-mode stochastic order should be stated and proved directly at ?? to avoid circularity.]

Theorem 2.12 (Gaussian fidelity converse). *For the displaced-thermal cloning problem $\Phi_z \mapsto \Psi_z$, no channel can have asymptotic flat-prior, and hence no channel can have minimax, squared fidelity exceeding $F_G(\gamma, p)$. More precisely, if $C_L \subset \mathbb{C}^R$ is the cube $|\Re z_{ij}|, |\Im z_{ij}| \leq L$, then*

$$\lim_{L \rightarrow \infty} \sup_{\Lambda} \int_{C_L} B(\Lambda(\Phi_z), \Psi_z) \frac{dz}{|C_L|} = B_G(\gamma, p). \quad (35)$$

Proof. The lower bound is achieved by the product of quantum-limited amplifiers. For the upper bound, ?? allows us to restrict to displacement-covariant channels. Covariance reduces the fidelity to $z = 0$. Phase twirling cannot decrease fidelity with the phase-invariant product thermal target, so the output may be taken diagonal in the joint number basis. Let

$$T_{\mathbf{k}} = \prod_{\alpha} (1 - q_{\alpha}) q_{\alpha}^{k_{\alpha}}, \quad P_{\mathbf{k}}^* = \prod_{\alpha} (1 - q_{\alpha}^{(\gamma)}) (q_{\alpha}^{(\gamma)})^{k_{\alpha}}.$$

The first-variation weight at P^* is

$$W_{\mathbf{k}} = \sqrt{T_{\mathbf{k}}/P_{\mathbf{k}}^*} = C \prod_{\alpha} \left(\frac{q_{\alpha}}{q_{\alpha}^{(\gamma)}} \right)^{k_{\alpha}/2},$$

a bounded product of decreasing one-mode weights. By ??, the expectation of W under any admissible covariant amplifier output is no larger than under P^* . Concavity of the Bhattacharyya functional $P \mapsto \sum_{\mathbf{k}} \sqrt{P_{\mathbf{k}} T_{\mathbf{k}}}$ gives the supporting-hyperplane bound

$$B(P, T) \leq B(P^*, T) + \frac{1}{2} \sum_{\mathbf{k}} W_{\mathbf{k}} (P_{\mathbf{k}} - P_{\mathbf{k}}^*) \leq B(P^*, T).$$

The value $B(P^*, T)$ is exactly $B_G(\gamma, p)$. This proves (?). □

3 Known-spectrum cloning

In this section the spectrum p is fixed and known. Only the orbit element g is unknown.

3.1 The known-spectrum Schur–PRV channel

Let $m = m_n$, $m/n \rightarrow \gamma > 1$. For a typical input Young diagram $\mu \vdash n$, define the admissible output set

$$\mathbf{A}(\mu) = \{v \vdash m : v \in T_m(p), \omega := v - \mu \text{ is dominant}\}. \quad (36)$$

If $\mu \in T_n(p)$ and $v \in T_m(p)$, then

$$(v_i - \mu_i) - (v_{i+1} - \mu_{i+1}) = (m - n)(p_i - p_{i+1}) + O(n^{2/3}) > 0$$

for all large n . Hence $P_{m,p}(\mathbf{A}(\mu)^c) = o(1)$, uniformly over typical μ . Define the Markov kernel

$$K_n(v|\mu) = \frac{P_{m,p}(v)}{P_{m,p}(\mathbf{A}(\mu))} \mathbf{1}_{v \in \mathbf{A}(\mu)}. \quad (37)$$

For atypical μ , choose any fallback kernel and channel.

For $v \in A(\mu)$, set $\omega = v - \mu$. As recalled in ??, the tensor product $\mathcal{H}_\mu \otimes \mathcal{H}_\omega$ contains the Cartan/PRV component of highest weight $\mu + \omega = v$. Let

$$V_{\mu,\omega} : \mathcal{H}_v \hookrightarrow \mathcal{H}_\mu \otimes \mathcal{H}_\omega \quad (38)$$

be the normalized intertwining isometry satisfying $V_{\mu,\omega} v_v = v_\mu \otimes v_\omega$ on highest-weight vectors. We call this the PRV inclusion, following the extremal-weight perspective on Cartan/PRV components [?, ?].

Definition 3.1 (Sector PRV channel). *The PRV channel from $\mathcal{H}_\mu$ to $\mathcal{H}_v$ is*

$$C_{\mu \rightarrow v}^{\text{PRV}}(X) = \frac{d_\mu}{d_v} V_{\mu,\omega}^* (X \otimes I_\omega) V_{\mu,\omega}, \quad d_\lambda = \dim \mathcal{H}_\lambda. \quad (39)$$

This formula is in the Schrodinger picture: X is an operator on $\mathcal{H}_\mu$, and the output is an operator on $\mathcal{H}_v$. To see complete positivity, choose an orthonormal basis $\{e_a\}$ of $\mathcal{H}_\omega$ and set

$$W_a = (I \otimes \langle e_a |) V_{\mu,\omega} : \mathcal{H}_v \rightarrow \mathcal{H}_\mu.$$

Then

$$C_{\mu \rightarrow v}^{\text{PRV}}(X) = \sum_a K_a X K_a^*, \quad K_a = \sqrt{d_\mu/d_v} W_a^*.$$

Trace preservation follows from Schur's lemma: $\text{Tr}_\omega(VV^*)$ commutes with the irreducible $U(d)$ -action on $\mathcal{H}_\mu$, hence equals $(d_v/d_\mu)I_\mu$, because its trace is d_v .

Definition 3.2 (Known-spectrum Schur-PRV cloner). *Given an input state on $(\mathbb{C}^d)^{\otimes n}$:*

- (i) *apply the Schur transform (??);*
- (ii) *measure $\mu \vdash n$;*
- (iii) *discard the input Specht register and keep the Weyl-sector marginal on $\mathcal{H}_\mu$; for the orbit inputs this is lossless, since the Specht factor in (??) is maximally mixed and carries no orbit information;*
- (iv) *sample v according to (??);*
- (v) *apply $C_{\mu \rightarrow v}^{\text{PRV}}$;*
- (vi) *prepare $I_{\mathcal{S}_v}/\dim \mathcal{S}_v$;*
- (vii) *apply the inverse Schur transform on m systems.*

The resulting channel is denoted $C_{n \rightarrow m}^{\text{PRV},p}$.

For the orbit input, the output in Schur form is

$$C_{n \rightarrow m}^{\text{PRV},p}(\rho_{p,g}^{\otimes n}) = \bigoplus_{v \vdash m} \left[\sum_{\mu \vdash n} P_{n,p}(\mu) K_n(v|\mu) C_{\mu \rightarrow v}^{\text{PRV}}(\rho_{p,g,\mu}) \right] \otimes \frac{I_{\mathcal{S}_v}}{\dim \mathcal{S}_v}. \quad (40)$$

For completeness, here is the same channel on an arbitrary input operator R written directly in Schur form. Let Π_μ be the projection onto $\mathcal{H}_\mu \otimes \mathcal{S}_\mu$, and define the unnormalized Weyl-sector marginal

$$R_\mu := \text{Tr}_{\mathcal{S}_\mu}(\Pi_\mu R \Pi_\mu) \in \mathcal{B}(\mathcal{H}_\mu). \quad (41)$$

Then, up to the harmless fallback choices on atypical or inadmissible sectors,

$$C_{n \rightarrow m}^{\text{PRV}, p}(R) = \bigoplus_{\nu \vdash m} \left[\sum_{\mu \vdash n} K_n(\nu|\mu) C_{\mu \rightarrow \nu}^{\text{PRV}}(R_\mu) \right] \otimes \frac{I_{\mathcal{S}_\nu}}{\dim \mathcal{S}_\nu}, \quad (42)$$

followed by the inverse Schur transform. Formula (??) is the specialization of (??) to $R = \rho_{p,g}^{\otimes n}$, for which $R_\mu = P_{n,p}(\mu) \rho_{p,g,\mu}$.

3.2 Thermal PRV sector limit

The full displaced-state PRV-to-amplifier contraction is stronger than what is needed for all-site fidelity. Since the global Schur–PRV channel is $U(d)$ -covariant, the fidelity with $\rho_{p,g}^{\otimes m}$ is independent of g . For achievability we may therefore set $g = I$. The necessary sector statement is purely thermal: it identifies the PRV output on $\rho_{p,\mu}$ with the thermal output of the least-noise amplifier.

Let

$$R_+ = \{(i, j) : 1 \leq i < j \leq d\}.$$

For $\alpha = (i, j) \in R_+$, write

$$F_\alpha = E_{ji}, \quad E_\alpha = E_{ij}, \quad \Delta_\alpha^\lambda = \lambda_i - \lambda_j, \quad q_\alpha = q_{ij} = p_j/p_i.$$

We fix once and for all a convex PBW ordering of the negative root vectors F_α . For a multi-index $\mathbf{r} = (r_\alpha)_{\alpha \in R_+}$, set

$$F^\mathbf{r} = \prod_{\alpha \in R_+}^< F_\alpha^{r_\alpha}, \quad |\mathbf{r}| = \sum_\alpha r_\alpha.$$

Lemma 3.3 (Heisenberg contraction of gap-rescaled root operators). *Fix $L < \infty$, and let $\lambda = \lambda^{(N)}$ range over a compact subcone of the strict Weyl chamber, so that $cN \leq \Delta_\alpha^\lambda \leq CN$ for all $\alpha \in R_+$. Put*

$$\widehat{F}_\alpha = F_\alpha / \sqrt{\Delta_\alpha^\lambda}, \quad \widehat{E}_\alpha = E_\alpha / \sqrt{\Delta_\alpha^\lambda}.$$

On every fixed excitation cutoff $|\mathbf{r}| \leq L$, uniformly over such λ ,

$$[\widehat{F}_\alpha, \widehat{F}_\beta] = O_L(N^{-1/2}), \quad [\widehat{E}_\alpha, \widehat{E}_\beta] = O_L(N^{-1/2}), \quad (43)$$

$$[\widehat{E}_\alpha, \widehat{F}_\beta] = \delta_{\alpha\beta} I + O_L(N^{-1/2}). \quad (44)$$

Consequently the rescaled generators converge on cutoffs to independent oscillator creation and annihilation operators indexed by R_+ .

Proof. In type A, $[F_\alpha, F_\beta] = N_{\alpha\beta} F_{\alpha+\beta}$, with $N_{\alpha\beta} \in \{0, \pm 1\}$, and the bracket is zero unless $\alpha + \beta$ is a root. If $\alpha + \beta$ is a root, then

$$[\widehat{F}_\alpha, \widehat{F}_\beta] = N_{\alpha\beta} \frac{\sqrt{\Delta_{\alpha+\beta}^\lambda}}{\sqrt{\Delta_\alpha^\lambda \Delta_\beta^\lambda}} \widehat{F}_{\alpha+\beta}.$$

On a fixed cutoff, $\widehat{F}_{\alpha+\beta}$ has uniformly bounded matrix elements, while the scalar prefactor is $O(N^{-1/2})$. This gives the first bound in (??); the second is its adjoint.

For $\alpha = \beta$, $[\widehat{E}_\alpha, \widehat{F}_\alpha] = H_\alpha / \Delta_\alpha^\lambda$. On a vector of weight $\lambda - \sum_\beta r_\beta \beta$ with $|\mathbf{r}| \leq L$, the Cartan element H_α acts by $\Delta_\alpha^\lambda + O_L(1)$. Hence $H_\alpha / \Delta_\alpha^\lambda = I + O_L(N^{-1})$ on the cutoff. For $\alpha \neq \beta$, $[E_\alpha, F_\beta]$ is either zero or a single root vector, and the same gap-counting argument gives $O_L(N^{-1/2})$. This proves (??). $\square$

Remark 3.4 (The same-weight monomial issue in type A_2). The order $N^{-1/2}$ is exactly what occurs in the basic same-weight example. In $\mathfrak{gl}_3$, the normalized composite-root vector $F_{13}v_\lambda/\sqrt{\Delta_{13}}$ and the normalized product $F_{12}F_{23}v_\lambda/\sqrt{\Delta_{12}\Delta_{23}}$ have inner product of order

$$-\sqrt{\frac{\Delta_{23}}{\Delta_{12}\Delta_{13}}} = O(N^{-1/2}),$$

up to the convention-dependent sign. Thus same-weight but distinct PBW monomials become orthogonal in the contraction limit.

Lemma 3.5 (Uniform PBW/Fock normal form). Fix $L < \infty$. Let $\lambda = \lambda^{(N)}$ range over dominant weights in a compact subcone of the strict Weyl chamber, so that

$$cN \leq \Delta_\alpha^\lambda \leq CN \quad (\alpha \in R_+)$$

for constants $0 < c < C < \infty$. Define

$$\xi_{\mathbf{r}}^\lambda = \prod_{\alpha \in R_+}^< \frac{F_\alpha^{r_\alpha}}{\sqrt{r_\alpha!} (\Delta_\alpha^\lambda)^{r_\alpha/2}} v_\lambda, \quad |\mathbf{r}| \leq L. \quad (45)$$

Then, uniformly over such λ ,

$$\langle \xi_{\mathbf{r}}^\lambda, \xi_{\mathbf{s}}^\lambda \rangle = \delta_{\mathbf{r},\mathbf{s}} + O_L(N^{-1/2}). \quad (46)$$

After Gram–Schmidt there are orthonormal vectors $e_{\mathbf{r}}^\lambda$ with

$$e_{\mathbf{r}}^\lambda = \xi_{\mathbf{r}}^\lambda + O_L(N^{-1/2}), \quad |\mathbf{r}| \leq L,$$

and the normalized root operators satisfy, on the cutoff $|\mathbf{r}| \leq L$,

$$\frac{F_\alpha}{\sqrt{\Delta_\alpha^\lambda}} e_{\mathbf{r}}^\lambda = \sqrt{r_\alpha + 1} e_{\mathbf{r}+\mathbf{e}_\alpha}^\lambda + O_L(N^{-1/2}), \quad (47)$$

$$\frac{E_\alpha}{\sqrt{\Delta_\alpha^\lambda}} e_{\mathbf{r}}^\lambda = \sqrt{r_\alpha} e_{\mathbf{r}-\mathbf{e}_\alpha}^\lambda + O_L(N^{-1/2}). \quad (48)$$

Proof. The inner products are Shapovalov-form matrix elements

$$\langle F^{\mathbf{r}} v_\lambda, F^{\mathbf{s}} v_\lambda \rangle = \langle v_\lambda, (F^{\mathbf{r}})^* F^{\mathbf{s}} v_\lambda \rangle.$$

Move all raising operators to the right using the commutation relations in $\mathfrak{gl}_d$. Only Cartan terms survive on the highest-weight vector. The top-degree part in the gaps occurs when every E_α is paired with the corresponding F_α , giving

$$\delta_{\mathbf{r},\mathbf{s}} \prod_{\alpha} r_\alpha! (\Delta_\alpha^\lambda)^{r_\alpha}.$$

Every commutator either lowers the total number of root factors or replaces a direct pairing by a composite-root expression; on the fixed cutoff it therefore has degree at most one less in the gaps. This also treats the only delicate possibility, namely distinct PBW monomials with the same total weight, such as the type A_2 relation between a composite root vector and a product of simple root vectors. In the associated graded calculation these are distinct PBW monomials; their overlap can only be produced by commutator terms and hence by lower-degree gap polynomials. Since all

gaps are comparable to N , the normalized error is $O_L(N^{-1/2})$, in fact often $O_L(N^{-1})$. This proves (??). Gram–Schmidt changes the cutoff vectors by the same order. The creation and annihilation formulas (??)–(??) follow by applying the same commutator estimate to $F_\alpha F^{\mathbf{r}} v_\lambda$ and to $E_\alpha F^{\mathbf{r}} v_\lambda$. The constants are uniform because L is fixed and the chamber constants c, C bound all denominators away from zero. $\square$

Lemma 3.6 (Thermal limit of a Weyl-sector Gibbs state). *Let $\lambda^{(N)}/N \rightarrow p$ with $p_1 > \dots > p_d > 0$. In the PBW/Fock realization of Lemma ??,*

$$\rho_{p,\lambda} = \frac{\pi_\lambda(\rho_p)}{s_\lambda(p)} \longrightarrow \bigotimes_{\alpha=(i,j) \in R_+} \tau_{q_\alpha} \quad (49)$$

in trace norm, where $\tau_q = (1 - q) \sum_{r \geq 0} q^r |r\rangle\langle r|$.

Proof. The vector $F^{\mathbf{r}} v_\lambda$ has weight

$$\lambda - \sum_{\alpha=(i,j)} r_\alpha (e_i - e_j).$$

Therefore

$$\pi_\lambda(\rho_p) F^{\mathbf{r}} v_\lambda = p^\lambda \prod_{\alpha=(i,j)} \left(\frac{p_j}{p_i} \right)^{r_\alpha} F^{\mathbf{r}} v_\lambda = p^\lambda \prod_{\alpha} q_\alpha^{r_\alpha} F^{\mathbf{r}} v_\lambda.$$

Because p lies in the strict chamber, the Weyl character formula gives, uniformly for λ/N in a compact set around p ,

$$s_\lambda(p) = p^\lambda \prod_{\alpha \in R_+} (1 - q_\alpha)^{-1} (1 + o(1)). \quad (50)$$

Indeed all non-identity Weyl terms are exponentially smaller, since permuting a strict decreasing p lowers the exponential weight. Combining this with Lemma ??, the matrix of $\rho_{p,\lambda}$ on every fixed excitation cutoff converges to the diagonal product distribution

$$\prod_{\alpha} (1 - q_\alpha) q_\alpha^{r_\alpha}.$$

The limiting product thermal state has an exponentially small tail outside $|\mathbf{r}| \leq L$, uniformly as $L \rightarrow \infty$. The same tail bound for $\rho_{p,\lambda}$ follows from (??) and the fact that all $q_\alpha < 1$. Taking first $N \rightarrow \infty$ on a fixed cutoff and then $L \rightarrow \infty$ proves trace-norm convergence. $\square$

Proposition 3.7 (Thermal PRV sector limit). *Let $\mu_n \vdash n$, $\nu_n \vdash m_n$, $m_n/n \rightarrow \gamma > 1$, satisfy*

$$\mu_n/n \rightarrow p, \quad \nu_n/m_n \rightarrow p,$$

and set $\omega_n = \nu_n - \mu_n$. Then ω_n is dominant for all large n . In the PBW/Fock realization of $\mathcal{H}_{\nu_n}$,

$$C_{\mu_n \rightarrow \nu_n}^{\text{PRV}}(\rho_{p,\mu_n}) \longrightarrow \bigotimes_{i < j} \tau_{q_{ij}^{(\gamma)}} \quad (51)$$

in trace norm, where

$$q_{ij}^{(\gamma)} = 1 - \frac{1 - q_{ij}}{\gamma}, \quad q_{ij} = p_j/p_i.$$

Consequently,

$$F\left(C_{\mu_n \rightarrow \nu_n}^{\text{PRV}}(\rho_{p,\mu_n}), \rho_{p,\nu_n}\right) \longrightarrow F_{\text{orb}}(\gamma, p). \quad (52)$$

Both convergences are uniform over all sequences (μ_n, ν_n) satisfying $\|\mu_n/n - p\|_\infty \leq \delta_n$ and $\|\nu_n/m_n - p\|_\infty \leq \delta_n$ for any fixed sequence $\delta_n \rightarrow 0$.

Proof. Dominance follows from

$$(\omega_i - \omega_{i+1}) = (v_i - v_{i+1}) - (\mu_i - \mu_{i+1}) = (m_n - n)(p_i - p_{i+1}) + o(n) > 0.$$

Let $V = V_{\mu, \omega} : \mathcal{H}_\nu \hookrightarrow \mathcal{H}_\mu \otimes \mathcal{H}_\omega$. The exact intertwining identity is

$$(\mathrm{d}\pi_\mu(Y) \otimes I + I \otimes \mathrm{d}\pi_\omega(Y))V = V \mathrm{d}\pi_\nu(Y) \quad (53)$$

for every $Y \in \mathfrak{gl}_d$. Taking $Y = F_\alpha$ gives

$$VF_\alpha^{(\nu)} = (F_\alpha^{(\mu)} \otimes I + I \otimes F_\alpha^{(\omega)})V. \quad (54)$$

Put

$$\eta_\alpha^{(n)} = \frac{\Delta_\alpha^\mu}{\Delta_\alpha^\nu}, \quad 1 - \eta_\alpha^{(n)} = \frac{\Delta_\alpha^\omega}{\Delta_\alpha^\nu}.$$

Since $\Delta_\alpha^\nu = \Delta_\alpha^\mu + \Delta_\alpha^\omega$ and the sectors are typical, $\eta_\alpha^{(n)} \rightarrow 1/\gamma$.

Apply (??) to PBW monomials and reorder into the fixed PBW order. By Lemma ??, commutator terms are $O_L(n^{-1/2})$ on every fixed excitation cutoff. Hence, uniformly for $|\mathbf{r}| \leq L$,

$$Ve_{\mathbf{r}}^\nu = \sum_{\mathbf{a} \leq \mathbf{r}} \prod_{\alpha \in R_+} \sqrt{\binom{r_\alpha}{a_\alpha}} (\eta_\alpha^{(n)})^{a_\alpha} (1 - \eta_\alpha^{(n)})^{r_\alpha - a_\alpha} e_{\mathbf{a}}^\mu \otimes e_{\mathbf{r}-\mathbf{a}}^\omega + O_L(n^{-1/2}). \quad (55)$$

This is the finite-cutoff beam-splitter binomial law, but we only use it at the thermal point.

Let

$$\sigma_{\mu, \nu} = C_{\mu \rightarrow \nu}^{\text{PRV}}(\rho_{p, \mu}).$$

For fixed $\mathbf{r}, \mathbf{s}$,

$$\langle e_{\mathbf{r}}^\nu, \sigma_{\mu, \nu} e_{\mathbf{s}}^\nu \rangle = \frac{d_\mu}{d_\nu} \langle Ve_{\mathbf{r}}^\nu, (\rho_{p, \mu} \otimes I) Ve_{\mathbf{s}}^\nu \rangle.$$

Using (??) and Lemma ??, off-diagonal entries tend to zero. For $\mathbf{r} = \mathbf{s}$, the Weyl dimension formula gives

$$\frac{d_\mu}{d_\nu} = \prod_{\alpha \in R_+} \frac{\Delta_\alpha^\mu + O(1)}{\Delta_\alpha^\nu + O(1)} = \prod_{\alpha} \eta_\alpha^{(n)} + o(1).$$

Therefore

$$\begin{aligned} \langle e_{\mathbf{r}}^\nu, \sigma_{\mu, \nu} e_{\mathbf{r}}^\nu \rangle &\rightarrow \prod_{\alpha} \eta_\alpha (1 - q_\alpha) \sum_{a_\alpha=0}^{r_\alpha} \binom{r_\alpha}{a_\alpha} (\eta_\alpha q_\alpha)^{a_\alpha} (1 - \eta_\alpha)^{r_\alpha - a_\alpha} \\ &= \prod_{\alpha} \eta_\alpha (1 - q_\alpha) (1 - \eta_\alpha + \eta_\alpha q_\alpha)^{r_\alpha}, \end{aligned}$$

where $\eta_\alpha = \lim \eta_\alpha^{(n)} = 1/\gamma$. Define

$$q'_\alpha = 1 - \eta_\alpha (1 - q_\alpha).$$

Then $1 - q'_\alpha = \eta_\alpha (1 - q_\alpha)$, and the last display is

$$\prod_{\alpha} (1 - q'_\alpha) (q'_\alpha)^{r_\alpha}.$$

Thus $\sigma_{\mu,v}$ converges on every fixed cutoff to $\bigotimes_{\alpha} \tau_{q'_{\alpha}}$. Since the limiting thermal state has exponentially small cutoff tail, and the same geometric bound follows for $\sigma_{\mu,v}$ from the displayed diagonal formula with $q'_{\alpha} < 1$, the convergence is in trace norm. Finally $q'_{\alpha} = 1 - (1 - q_{\alpha})/\gamma$, proving (??). Combining this with Lemma ?? for v_n and the explicit formula for thermal fidelity gives (??). For the uniformity claim, observe that every estimate above depends only on the cutoff L , on the chamber constants c, C of Lemma ??, and on the ratios $\eta_{\alpha}^{(n)}$. Under $\|\mu_n/n - p\|_{\infty} \leq \delta_n$ and $\|v_n/m_n - p\|_{\infty} \leq \delta_n$, the gaps satisfy $\Delta_{\alpha}^{\mu_n} = n(p_i - p_j) + O(n\delta_n)$ and $\Delta_{\alpha}^{v_n} = m_n(p_i - p_j) + O(m_n\delta_n)$, so $\eta_{\alpha}^{(n)} \rightarrow 1/\gamma$ uniformly over such sequences, and all limits above are attained uniformly. $\square$

Remark 3.8 (Optional stronger channel statement). *The stronger statement that the PRV channel converges on displaced local states to the quantum-limited amplifier follows from the same PBW normal form plus the action of local Lie algebra perturbations. It is useful conceptually, but the known-spectrum all-site fidelity theorem only uses Proposition ??.*

3.3 Known-spectrum achievability

Theorem 3.9 (Known-spectrum achievability). *The Schur–PRV cloner satisfies*

$$\lim_{n \rightarrow \infty} \inf_{g \in \mathbf{U}(d)} F\left(C_{n \rightarrow m}^{\text{PRV},p}(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m}\right) = F_{\text{orb}}(\gamma, p). \quad (56)$$

Proof. The channel is $\mathbf{U}(d)$ -covariant, so the fidelity is independent of g . We take $g = I$. Let

$$Q_m(v) = \sum_{\mu} P_{n,p}(\mu) K_n(v|\mu)$$

be the output Young-label distribution. By Lemma ?? and the dominance estimate preceding (??),

$$\|Q_m - P_{m,p}\|_{\text{TV}} = o(1).$$

Indeed, for typical μ , the conditional law $K_n(\cdot|\mu)$ differs from $P_{m,p}$ by $o(1)$ in total variation; atypical μ 's have $o(1)$ probability.

The target state is

$$\rho_p^{\otimes m} = \bigoplus_{v \vdash m} P_{m,p}(v) \rho_{p,v} \otimes \frac{I_{\mathcal{S}_v}}{\dim \mathcal{S}_v}.$$

For every typical admissible pair (μ, v) , Proposition ?? implies

$$B\left(C_{\mu \rightarrow v}^{\text{PRV}}(\rho_{p,\mu}), \rho_{p,v}\right) \rightarrow B_{\text{orb}}(\gamma, p)$$

uniformly. The Specht factors agree exactly.

Using the block formula (??), and then concavity of root fidelity inside each output block,

$$\begin{aligned} & B\left(C_{n \rightarrow m}^{\text{PRV},p}(\rho_p^{\otimes n}), \rho_p^{\otimes m}\right) \\ & \geq \sum_v \sqrt{Q_m(v) P_{m,p}(v)} \sum_{\mu} w(\mu|v) B\left(C_{\mu \rightarrow v}^{\text{PRV}}(\rho_{p,\mu}), \rho_{p,v}\right), \end{aligned}$$

where $w(\mu|v)$ is the posterior weight of μ given v in the protocol. The atypical/admissibility complement contributes $o(1)$, while

$$\sum_v \sqrt{Q_m(v) P_{m,p}(v)} = 1 - o(1).$$

Thus the liminf of the root fidelity is at least $B_{\text{orb}}(\gamma, p)$. Conversely, the LAN converse in ?? below gives the matching upper bound for all channels, including the Schur–PRV channel. Squaring proves (??). $\square$

3.4 Known-spectrum global converse

Theorem 3.10 (Known-spectrum global converse). *For arbitrary channels*

$$\mathcal{M}_n : \mathcal{B}((\mathbb{C}^d)^{\otimes n}) \rightarrow \mathcal{B}((\mathbb{C}^d)^{\otimes m_n}),$$

$$\limsup_{n \rightarrow \infty} \sup_{\mathcal{M}_n} \inf_{g \in \text{U}(d)} F\left(\mathcal{M}_n(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}\right) \leq F_{\text{orb}}(\gamma, p). \quad (57)$$

Proof. It is enough to prove the root-fidelity statement. Suppose, for contradiction, that for some $\varepsilon > 0$, a subsequence and channels $\mathcal{M}_n$ satisfy

$$\inf_{g \in \text{U}(d)} B\left(\mathcal{M}_n(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}\right) \geq B_{\text{orb}}(\gamma, p) + 3\varepsilon. \quad (58)$$

Fix a cube $C_L \subset \mathbb{C}^R$. For every $z \in C_L$, the local state $\rho_{p,g_{n,z}}^{\otimes n}$ belongs to the original orbit family, hence (??) applies.

Let T_{m_n} be the output LAN channel and S_n the inverse input LAN channel from ?. Define the induced oscillator channel

$$\Gamma_n = T_{m_n} \circ \mathcal{M}_n \circ S_n.$$

By the LAN transfer principle (??), uniformly for $z \in C_L$,

$$\begin{aligned} B(\Gamma_n(\Phi_z), \Psi_z) &\geq B\left(\mathcal{M}_n(\rho_{p,g_{n,z}}^{\otimes n}), \rho_{p,g_{n,z}}^{\otimes m_n}\right) - o(1) \\ &\geq B_{\text{orb}}(\gamma, p) + 2\varepsilon \end{aligned}$$

for all large n . The first line uses (??), monotonicity under T_{m_n} , (??), and Lemma ?. Therefore

$$\int_{C_L} B(\Gamma_n(\Phi_z), \Psi_z) \frac{dz}{|C_L|} \geq B_{\text{orb}}(\gamma, p) + 2\varepsilon.$$

Since Γ_n is an arbitrary oscillator channel, this contradicts the flat-prior Gaussian bound (??) after choosing L large enough. Hence no such finite-dimensional cloner exists. Squaring the root-fidelity bound proves (??). $\square$

Combining ???? gives the known-spectrum optimality theorem.

Corollary 3.11 (Known-spectrum asymptotic optimum).

$$\limsup_{n \rightarrow \infty} \inf_{\mathcal{M}_n} \inf_{g \in \text{U}(d)} F\left(\mathcal{M}_n(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}\right) = F_{\text{orb}}(\gamma, p),$$

and the value is achieved by the Schur–PRV cloner.

4 Unknown-spectrum adaptive cloning

We now assume the spectrum is not known. The model is the full nondegenerate family

$$\rho_{p,g} = U(g) \text{diag}(p) U(g)^*, \quad p \in \Delta_d^\circ.$$

The sharp statement is local at a fixed p , uniformly on compact subsets of the nondegenerate chamber. The additional loss relative to known spectrum comes from the classical LAN component.

4.1 Adaptive Schur-PRV protocol

The protocol again begins with the Schur transform and a measurement of $\mu \vdash n$. Now p is not known, so one does not sample v from $P_{m,p}$. Instead choose a rounded dilation

$$R_{n,m}(\mu) = v \vdash m, \quad \left\| v - \frac{m}{n}\mu \right\|_\infty \leq C_d, \quad (59)$$

with $v_1 \geq \dots \geq v_d \geq 0$. Such a rounding map exists by distributing the integer rounding errors while preserving the order. If μ/n lies in a compact nondegenerate chamber and $\gamma > 1$, then $\omega = v - \mu$ is dominant for all large n , because

$$(\omega_i - \omega_{i+1}) = (\gamma_n - 1)(\mu_i - \mu_{i+1}) + O(1) > 0.$$

Then apply the same sector PRV channel $C_{\mu \rightarrow v}^{\text{PRV}}$, prepare $I_{S_v}/\dim S_v$, and invert the Schur transform. Denote the adaptive channel by $C_{n \rightarrow m}^{\text{adPRV}}$.

The important point is that the PRV step itself does not need p . It only needs the measured sector μ and the chosen output sector v .

4.2 The classical Young-label factor

Let $\mathbf{H} = \{x \in \mathbb{R}^d : \sum_i x_i = 0\}$ and $V_p = \text{diag}(p) - pp^T$ restricted to $\mathbf{H}$. The Schur label satisfies a central-limit theorem compatible with ??:

$$X_n := \frac{\mu - np}{\sqrt{n}} \Rightarrow X \sim \mathbf{N}_{0,V_p}. \quad (60)$$

For a local spectrum perturbation $p + h/\sqrt{n}$, the limit is $\mathbf{N}_{h,V_p}$.

The adaptive dilation $v = R_{n,m}(\mu)$ gives

$$\frac{v - mp}{\sqrt{m}} = \frac{(m/n)\mu - mp}{\sqrt{m}} + o(1) = \sqrt{\frac{m}{n}} \frac{\mu - np}{\sqrt{n}} + o(1) \Rightarrow \mathbf{N}_{0,\gamma V_p}. \quad (61)$$

For the target m -copy state, the Young label has limit $\mathbf{N}_{0,V_p}$. Thus the classical squared fidelity contribution is

$$F_{\text{cl}}(\gamma, d) = F(\mathbf{N}_{0,\gamma V_p}, \mathbf{N}_{0,V_p}). \quad (62)$$

Lemma 4.1 (Classical Gaussian factor). *For any positive definite covariance matrix V on a k -dimensional real vector space,*

$$F(\mathbf{N}_{0,\gamma V}, \mathbf{N}_{0,V}) = \left(\frac{2\sqrt{\gamma}}{1 + \gamma} \right)^k. \quad (63)$$

In the spectrum problem $k = d - 1$, so (??) equals

$$F_{\text{cl}}(\gamma, d) = \left(\frac{2\sqrt{\gamma}}{1 + \gamma} \right)^{d-1}. \quad (64)$$

Proof. The squared Hellinger fidelity between centered Gaussians with covariances Σ_1, Σ_2 is

$$F = \frac{\det(\Sigma_1)^{1/2} \det(\Sigma_2)^{1/2}}{\det((\Sigma_1 + \Sigma_2)/2)}.$$

Taking $\Sigma_1 = \gamma V$, $\Sigma_2 = V$, and $\dim V = k$, gives

$$F = \frac{\gamma^{k/2} \det V}{((1 + \gamma)/2)^k \det V} = \left(\frac{2\sqrt{\gamma}}{1 + \gamma} \right)^k.$$

□

Lemma 4.2 (Uniform adaptive Young-label factor). *Fix $\kappa > 0$, $L < \infty$, and $m_n/n \rightarrow \gamma > 1$. For $p \in \Delta_{d,\kappa}$ and $h \in \mathcal{H}$ with $\|h\| \leq L$, let μ be the Schur label drawn from $P_{n,p+h/\sqrt{n}}$, and let $v = R_{n,m}(\mu)$ be the rounded dilation (??). Then the Young-label experiment converges uniformly to the classical Gaussian amplification experiment*

$$X \sim \mathbf{N}_{h,V_p}, \quad Y = \sqrt{\gamma}X \sim \mathbf{N}_{\sqrt{\gamma}h,\gamma V_p},$$

whereas the target m -copy Young-label experiment converges uniformly to $\mathbf{N}_{\sqrt{\gamma}h,V_p}$. Consequently the Young-label contribution to squared fidelity converges uniformly to

$$F_{\text{cl}}(\gamma, d) = \left(\frac{2\sqrt{\gamma}}{1 + \gamma} \right)^{d-1}.$$

Proof. The Schur label is the classical component of the full mixed-state LAN model. By ??, this classical component satisfies a local central limit theorem uniformly on $\Delta_{d,\kappa}$ and bounded h . The rounding map obeys

$$\frac{R_{n,m}(\mu) - mp}{\sqrt{m}} = \sqrt{\frac{m}{n}} \frac{\mu - np}{\sqrt{n}} + o(1),$$

with an $o(1)$ term uniform in the stated compact set because the rounding error is $O_d(1)$. Hellinger affinity is continuous under total-variation convergence of the local experiments, and the Gaussian value is given by ?. This proves the uniform convergence of the classical factor. □

A tempting but suboptimal alternative is to set $\hat{p} = \mu/n$ and then sample $v \sim P_{m,\hat{p}}$. At LAN scale this gives

$$\frac{v - mp}{\sqrt{m}} \approx \sqrt{\gamma}X + Y_0, \quad Y_0 \sim \mathbf{N}_{0,V_p},$$

so the covariance is $(\gamma + 1)V_p$, not γV_p . Its classical factor would be $(2\sqrt{\gamma + 1}/(\gamma + 2))^{d-1}$, which is worse.

4.3 Unknown-spectrum theorem

Define

$$F_{\text{univ}}(\gamma, p) = F_{\text{cl}}(\gamma, d) F_{\text{orb}}(\gamma, p) = \left(\frac{2\sqrt{\gamma}}{1 + \gamma} \right)^{d-1} \prod_{i < j} \frac{(1 - q_{ij})(1 - q_{ij}^{(\gamma)})}{(1 - \sqrt{q_{ij}q_{ij}^{(\gamma)}})^2}. \quad (65)$$

Proposition 4.3 (Classical Gaussian amplification converse). *For the classical Gaussian shift problem*

$$X \sim \mathbf{N}_{h,V_p} \longmapsto Y \sim \mathbf{N}_{\sqrt{\gamma}h,V_p}, \quad h \in \mathcal{H},$$

the flat-prior/minimax squared Hellinger fidelity is $F_{\text{cl}}(\gamma, d)$. It is achieved by the deterministic Markov kernel $Y = \sqrt{\gamma}X$.

Proof. The deterministic kernel gives output law $N_{\sqrt{\gamma}h, \gamma V_p}$, whose squared Hellinger fidelity with the target is $F_{\text{cl}}(\gamma, d)$ by ?? . Conversely, by the Hunt–Stein reduction of ?? , applied to the amenable translation group H , a flat-prior optimizer may be assumed translation covariant. A translation-covariant Markov kernel has the form $Y = \sqrt{\gamma}X + Z$, with Z independent of the input. At the origin the output density is $f_\eta := N_{0, \gamma V_p} * \eta$, where η is the law of Z , and the target density is $g := N_{0, V_p}$.

Write $f_0 := N_{0, \gamma V_p}$ and apply the variational formula of ?? with the witness $a_* := \sqrt{g/f_0}$, the optimizer for the pair (f_0, g) ; by construction

$$\int a_*^{-1} g \, dy = \int \sqrt{f_0 g} \, dy = B_{\text{cl}}(\gamma, d).$$

Since $\gamma > 1$, the witness a_* is a centered Gaussian function, and f_0 is a centered Gaussian density, so the convolution

$$(a_* * f_0)(r) = \int a_*(y) f_0(y - r) \, dy$$

is again a centered Gaussian function of r , maximized at $r = 0$, with maximal value $(a_* * f_0)(0) = \int \sqrt{f_0 g} \, dy = B_{\text{cl}}(\gamma, d)$. Hence

$$\int a_* f_\eta \, dy = \int_H (a_* * f_0)(r) \eta(dr) \leq B_{\text{cl}}(\gamma, d).$$

The variational formula then gives

$$B(f_\eta, g) \leq \frac{1}{2} \left\{ \int a_* f_\eta \, dy + \int a_*^{-1} g \, dy \right\} \leq B_{\text{cl}}(\gamma, d),$$

with equality for $\eta = \delta_0$. Squaring shows that the deterministic kernel is optimal, with value $F_{\text{cl}}(\gamma, d)$. This argument is the $R = 0$ specialization of the classical-factor bound in the hybrid converse of ?? . $\square$

Lemma 4.4 (Variational form of root fidelity). *For faithful normal states ρ, τ ,*

$$B(\rho, \tau) = \inf_{A>0} \frac{1}{2} \{ \text{Tr}(\rho A) + \text{Tr}(\tau A^{-1}) \}.$$

For commuting classical states this is the scalar identity $\sqrt{xy} = \inf_{a>0} (ax + y/a)/2$ integrated over the sample space.

Proof. This is the standard operator arithmetic–geometric mean variational formula for Uhlmann root fidelity. In the commuting case it reduces pointwise to the scalar formula. $\square$

Theorem 4.5 (Hybrid classical–quantum Gaussian product converse). *For the hybrid LAN problem*

$$N_{h, V_p} \otimes \Phi_z \longmapsto N_{\sqrt{\gamma}h, V_p} \otimes \Psi_z,$$

the flat-prior/minimax optimal squared fidelity is

$$F_{\text{cl}}(\gamma, d) F_{\text{orb}}(\gamma, p).$$

The optimum is achieved by the product of the deterministic classical amplifier $X \mapsto \sqrt{\gamma}X$ and the product least-noise oscillator amplifier.

Proof. The lower bound is the product channel. The upper bound is proved in ??: the amenable Hunt–Stein reduction and a per-mode phase average reduce the flat-prior problem to an exactly covariant, phase-covariant hybrid channel; ?? gives the residual-instrument decomposition; ?? supplies the linear witness at the least-noise product output; and the tensorized one-mode conic order, ??, bounds each covariant oscillator branch, including trace-nonincreasing branches. Thus correlations between the classical residual and the oscillator noise cannot improve the product fidelity. $\square$

Theorem 4.6 (Unknown-spectrum local achievability). *Fix a compact local parameter set $K \subseteq \mathcal{H} \times \mathbb{C}^R$. For $(h, z) \in K$, put*

$$p_{n,h} = p + h/\sqrt{n}, \quad g_{n,z} = \exp(X(z)/\sqrt{n}),$$

and define

$$\rho_{n,h,z} = (\rho_{p_{n,h},g_{n,z}})^{\otimes n}, \quad \sigma_{m,h,z} = (\rho_{p_{n,h},g_{n,z}})^{\otimes m_n}.$$

Then the adaptive Schur–PRV channel satisfies

$$\liminf_{n \rightarrow \infty} \inf_{(h,z) \in K} F\left(C_{n \rightarrow m}^{\text{adPRV}}(\rho_{n,h,z}), \sigma_{m,h,z}\right) \geq F_{\text{univ}}(\gamma, p). \quad (66)$$

Proof. By ??, the Young-label component of the adaptive protocol converges uniformly on K to the deterministic classical amplifier $X \mapsto \sqrt{\gamma}X$, hence contributes $F_{\text{cl}}(\gamma, d)$. The channel is $U(d)$ -covariant, so for each local spectrum $p_{n,h}$ the fidelity is independent of the orbit coordinate z ; thus the PRV sector fidelity may be evaluated at $z = 0$. Conditional on typical labels, $v = R_{n,m}(\mu)$ satisfies $\Delta_{ij}^v/\Delta_{ij}^\mu \rightarrow \gamma$, so ?? gives the orbit factor $F_{\text{orb}}(\gamma, p)$, uniformly on compact LAN windows. Atypical Young labels have probability $o(1)$, and the rounding error in (??) is $o(\sqrt{n})$, hence invisible at LAN scale. Multiplying the root-fidelity factors and squaring proves (??). $\square$

For the converse, a fixed compact local window is not least favorable: on a singleton one can prepare the target and get fidelity one. The correct local minimax converse uses expanding windows. Let

$$C_L = \{(h, z) \in \mathcal{H} \times \mathbb{C}^R : \|h\| \leq L, |\Re z_\alpha|, |\Im z_\alpha| \leq L \text{ for all } \alpha\}.$$

Theorem 4.7 (Unknown-spectrum expanding-window local converse). *For the full-rank nondegenerate LAN model of ??, one has*

$$\lim_{L \rightarrow \infty} \limsup_{n \rightarrow \infty} \sup_{\mathcal{M}_n} \inf_{(h,z) \in C_L} F(\mathcal{M}_n(\rho_{n,h,z}), \sigma_{m,h,z}) \leq F_{\text{univ}}(\gamma, p). \quad (67)$$

Together with ??, this identifies $F_{\text{univ}}(\gamma, p)$ as the local asymptotic minimax value in the expanding-window LAN sense.

Proof. Suppose the root-fidelity version of (??) fails by 3ϵ . Then, for arbitrarily large L , there is a subsequence and finite-dimensional channels $\mathcal{M}_n$ whose fidelity is at least $B_{\text{univ}}(\gamma, p) + 3\epsilon$ uniformly on C_L . Compose $\mathcal{M}_n$ with the inverse input LAN channel and the output LAN channel from ??. The trace-norm-to-fidelity transfer in ?? gives a hybrid Gaussian channel with root fidelity at least $B_{\text{univ}}(\gamma, p) + 2\epsilon$ uniformly on C_L . Averaging over the uniform prior on C_L gives a flat-prior hybrid Gaussian cloner beating ??. Letting $L \rightarrow \infty$ yields a contradiction. Squaring gives (??). $\square$

The expanding-window local converse implies a genuine global worst-case statement on compact subsets of the strict Weyl chamber. The needed uniformity is supplied by ???????.

Corollary 4.8 (Compact-spectrum global worst-case value). *Fix $\kappa > 0$ and let*

$$\Delta_{d,\kappa} = \{p : p_d \geq \kappa, p_i - p_{i+1} \geq \kappa, \sum_i p_i = 1\}.$$

Then

$$\lim_{n \rightarrow \infty} \sup_{\mathcal{M}_n} \inf_{\substack{p \in \Delta_{d,\kappa} \\ g \in \mathcal{U}(d)}} F(\mathcal{M}_n(\rho_{p,g}^{\otimes n}), \rho_{p,g}^{\otimes m_n}) = \inf_{p \in \Delta_{d,\kappa}} F_{\text{univ}}(\gamma, p). \quad (68)$$

The adaptive Schur–PRV channel achieves the lower bound.

Proof. Achievability follows from ??, the uniform estimates ????????, and continuity of $p \mapsto F_{\text{univ}}(\gamma, p)$ on $\Delta_{d,\kappa}$. For the converse, suppose a global channel beats the right-hand side by 3ϵ . Choose $p_0 \in \Delta_{d,\kappa}$ with

$$F_{\text{univ}}(\gamma, p_0) \leq \inf_{p \in \Delta_{d,\kappa}} F_{\text{univ}}(\gamma, p) + \epsilon.$$

For every fixed L , the local perturbations $p_0 + h/\sqrt{n}$ with $(h, z) \in C_L$ remain in $\Delta_{d,\kappa}$ for all sufficiently large n . The assumed global worst-case bound therefore gives a local expanding-window cloner beating $F_{\text{univ}}(\gamma, p_0)$, contradicting ??. $\square$

A global worst-case statement over the entire full-rank simplex without a gap condition is not covered by (??): as p approaches degeneracy or the boundary, the LAN model changes and the formula $F_{\text{univ}}(\gamma, p)$ need not remain the correct local value.

Remark 4.9 (Known side information at intermediate precision). *If the spectrum is supplied externally to accuracy $o(n^{-1/2})$, the classical penalty disappears and the known-spectrum value F_{orb} is recovered. If the supplied accuracy is only $O(n^{-1/2})$, the classical LAN problem changes by replacing the input covariance V_p with the posterior covariance after incorporating that side information.*

5 Comparison with existing mixed-state cloning and broadcasting results

Broadcasting versus cloning. The present work is about cloning in the global product sense: the target is $\rho^{\otimes m}$. Broadcasting is weaker: it requires only that selected output marginals reproduce the input state. A perfect broadcaster can have strong correlations among the outputs, and therefore may have poor global fidelity with $\rho^{\otimes m}$. This distinction is essential for interpreting the mixed-state literature. The single-copy no-broadcasting theorem rules out exact broadcasting of noncommuting families from one input copy [?]. With several input copies, however, exact marginal broadcasting can be possible. In particular, [?] showed that the no-broadcasting theorem does not extend directly to more than one input copy for mixed qubits and introduced superbroadcasting, where the output one-site marginals can even be purer than the input for sufficiently many input copies. Such purifying marginal maps can be degraded by local depolarization to exact broadcasting, and exact broadcasting was also constructed explicitly for displaced thermal oscillator states with known thermal photon number in [?]. The review/extension [?] states explicitly that for sufficiently many input copies ideal broadcasting of noncommuting mixed states is possible, and the universal and phase-covariant maps are developed further in [?]. These results are not contradicted by no-cloning, because they do not produce an uncorrelated product $\rho^{\otimes m}$.

Earlier mixed-state cloning bounds. Rastegin studied global-fidelity, relative-error, and state-dependent limits for mixed-state cloning [?, ?, ?]. Fan and collaborators constructed mixed-state cloners in symmetric subspaces and for identical mixed qubits [?, ?]; Kazakov studied partial cloning and mixed-qubit copying in another restricted setting [?]. These works are useful benchmarks, but they do not provide a sharp arbitrary-dimensional proportional-output all-site asymptotic formula. The Gaussian setting is an exception in infinite dimension: Guta–Matsumoto solved displaced mixed Gaussian cloning via least-noise amplification [?]. Our finite-dimensional converse reduces to that theorem through LAN.

Random purification plus pure-state cloning. The coherent-inference construction of [?] treats rank- r mixed states by first applying a random-purification channel and then the optimal pure-state cloner in dimension dr . Tracing out the purifying registers gives mixed-state clones. Their stated all-site guarantee has the pure-state-cloning form

$$F_{\text{RP}}(n, m; d, r) \geq \frac{\binom{n+dr-1}{dr-1}}{\binom{m+dr-1}{dr-1}}. \quad (69)$$

With $m/n \rightarrow \gamma$,

$$\frac{\binom{n+dr-1}{dr-1}}{\binom{m+dr-1}{dr-1}} = \prod_{a=1}^{dr-1} \frac{n+a}{m+a} \rightarrow \gamma^{-(dr-1)}. \quad (70)$$

For full-rank states, $r = d$, this is $\gamma^{-(d^2-1)}$.

Our full-rank fixed-spectrum orbit problem has

$$F_{\text{orb}}(\gamma, p) = \prod_{i < j} f_{\gamma}(q_{ij}),$$

where

$$f_{\gamma}(q) := \left(\frac{\sqrt{\gamma} + \sqrt{q(\gamma - 1 + q)}}{\gamma + q} \right)^2.$$

We now claim that for every $q > 0$ and $\gamma > 1$,

$$f_{\gamma}(q) > \frac{1}{\gamma}. \quad (71)$$

Let us first check why each one-root factor is larger than $1/\gamma$. Since the denominator $\gamma + q$ is positive, the comparison $f_{\gamma}(q) > 1/\gamma$ is equivalent, after taking positive square roots, to

$$\sqrt{\gamma} + \sqrt{q(\gamma - 1 + q)} > \frac{\gamma + q}{\sqrt{\gamma}} = \sqrt{\gamma} + \frac{q}{\sqrt{\gamma}}.$$

After cancelling $\sqrt{\gamma}$, this becomes

$$\sqrt{q(\gamma - 1 + q)} > \frac{q}{\sqrt{\gamma}}.$$

Both sides are nonnegative, and squaring gives

$$q(\gamma - 1 + q) > \frac{q^2}{\gamma},$$

or equivalently

$$q(\gamma - 1) \left(1 + \frac{q}{\gamma}\right) > 0.$$

This last inequality is strict for $q > 0$ and $\gamma > 1$. Hence, for every $q > 0$ and $\gamma > 1$, we have $f_\gamma(q) > \frac{1}{\gamma}$, as claimed.

Therefore, with $R = d(d - 1)/2$,

$$F_{\text{orb}}(\gamma, p) > \gamma^{-R}. \quad (72)$$

Since $R < d^2 - 1$ for $d \geq 2$,

$$F_{\text{orb}}(\gamma, p) > \gamma^{-R} > \gamma^{-(d^2-1)}. \quad (73)$$

Thus the known-spectrum Schur-PRV optimum is strictly larger than the full-rank random-purification guarantee for all $\gamma > 1$.

For unknown spectrum,

$$F_{\text{univ}}(\gamma, p) = F_{\text{cl}}(\gamma, d)F_{\text{orb}}(\gamma, p).$$

Since

$$\frac{2\sqrt{\gamma}}{1 + \gamma} > \gamma^{-1/2} \quad (\gamma > 1),$$

we have

$$F_{\text{univ}}(\gamma, p) > \gamma^{-(d-1)/2} \gamma^{-d(d-1)/2} = \gamma^{-(d^2-1)/2} > \gamma^{-(d^2-1)}. \quad (74)$$

Again, this compares our sharp local full-rank asymptotics to the universal purify-and-clone guarantee, not to a proven exact fidelity formula for that specific channel restricted to a fixed orbit.

5.1 Large-gamma and near-identity expansions

The expansion in powers of $1/\gamma$ is only a way to compare high expansion ratios. It is not used in the asymptotic proof, where $\gamma > 1$ is fixed and the small parameter is $1/n$. When γ is close to one, the natural small parameter is $\epsilon = \gamma - 1$, and the exact formulas above should be used.

For fixed p , as $\gamma \rightarrow \infty$,

$$F_{\text{orb}}(\gamma, p) = \gamma^{-d(d-1)/2} \prod_{i < j} \left(1 + \sqrt{\frac{p_j}{p_i}}\right)^2 (1 + O(\gamma^{-1})). \quad (75)$$

The product prefactor matters for the sharp leading constant, but not for the exponent. The inverse ratio $(1 + \sqrt{p_i/p_j})^2$ would be incorrect unless accompanied by the missing factor p_j/p_i for each root.

Near $\gamma = 1$, write $\gamma = 1 + \epsilon$. For each root,

$$f_{1+\epsilon}(q) = 1 - \frac{\epsilon^2}{4q} + O(\epsilon^3),$$

so the known-spectrum infidelity is quadratic in $\gamma - 1$. By contrast, the pure-cloner dimension-ratio guarantee $\gamma^{-(d^2-1)}$ has a linear loss

$$(1 + \epsilon)^{-(d^2-1)} = 1 - (d^2 - 1)\epsilon + O(\epsilon^2).$$

6 Outlook and discussion

The results identify the asymptotic global fidelity for full-rank nondegenerate mixed-state cloning in the proportional regime. The known-spectrum problem is governed by the root-mode quantum amplifier, whereas the unknown-spectrum problem contains, in addition, a classical Gaussian amplification factor for the Young-label/spectral LAN component. The Schur–PRV construction realizes the quantum amplifier directly at the level of Weyl sectors, and the adaptive rounded-dilation rule for Young diagrams realizes the classical amplifier.

Several extensions remain natural. First, one can ask for quantitative finite- n estimates. This would require explicit constants in Keyl–Werner concentration, quantitative LAN bounds, and quantitative versions of the PBW/Fock approximation. Second, degenerate spectra and rank-deficient states require separate analysis: degeneracies introduce stabilizers and collapse some root modes, while boundary points of the simplex lead to non-Gaussian or constrained LAN limits. Third, it would be useful to understand whether the PRV sector map is exactly optimal at finite sector sizes, beyond its asymptotic optimality for the global cloning problem. Finally, implementation of the protocol requires efficient Schur transforms and efficient realization of the relevant Cartan/PRV intertwiners.

[Internal note: if targeting a shorter journal version, this outlook can be reduced to one paragraph. Circuit implementation, finite-sector optimality, and degenerate spectra are natural follow-up projects, not hypotheses of the present theorem.]

[Internal note: former appendices A–C have been incorporated into the main text. The proof-status appendix has been removed; its role is now only this removable comment. The theorem statements are asymptotic and do not claim finite- n rates or boundary/degenerate-spectrum formulas.]

A The hybrid product converse: residual instruments and conic order

This appendix proves the hybrid classical–quantum Gaussian product converse used in ?? . The proof combines the Hunt–Stein reduction for the amenable product group, a per-mode phase average, a residual-instrument decomposition of exactly covariant hybrid channels, and the tensorized one-mode conic order from ?? .

Lemma A.1 (Residual-instrument form of covariant hybrid channels). *Let Λ be a normal channel from the hybrid classical–quantum input model to the hybrid output model, exactly covariant under $G = \mathbb{H} \times \mathbb{C}^R$. The input action translates the classical variable by s and displaces the oscillator by w ; the output action translates the classical variable by $\sqrt{\gamma}s$ and displaces the oscillator by $\sqrt{\gamma}w$. Then there is a Borel quantum instrument $E \mapsto \mathcal{I}_E$ on $\mathbb{H}$, whose values are trace-nonincreasing CP maps on the oscillators and are displacement covariant with gain $\sqrt{\gamma}$, such that for every bounded measurable test function a and every bounded oscillator observable M , the output Ξ of Λ at the centered input $\mathbf{N}_{0,V_p} \otimes \Phi_0$ satisfies*

$$\int a(y) \operatorname{Tr}[M \Xi(dy)] = \int_{\mathbb{H}} \left(\int a(y) \mathbf{N}_{r,\gamma V_p}(dy) \right) \operatorname{Tr}[M \mathcal{I}(dr)(\Phi_0)]. \quad (76)$$

Equivalently, in measure notation,

$$\Xi(dy) = \int_{\mathbb{H}} \mathbf{N}_{r,\gamma V_p}(dy) \mathcal{I}(dr)(\Phi_0), \quad \int_{\mathbb{H}} \operatorname{Tr}[\mathcal{I}(dr)(\Phi_0)] = 1. \quad (77)$$

Proof. We use the standard classical–quantum Markov-kernel formulation; the existence of a Borel-measurable family of conditional instruments below is the usual disintegration of a normal channel with classical input and classical–quantum output, see e.g. the treatment of covariant instruments in [?]. For a classical input point $x \in \mathsf{H}$, the channel induces a quantum instrument $E \mapsto \Lambda_E^x$ on the oscillator system, where $E \subseteq \mathsf{H}$ is a Borel output event. Product covariance gives, for all Borel E , classical shifts s , oscillator displacements w , and oscillator trace-class inputs X ,

$$\Lambda_{E+\sqrt{\gamma}s}^{x+s}(D(w)XD(w)^*) = D(\sqrt{\gamma}w)\Lambda_E^x(X)D(\sqrt{\gamma}w)^*. \quad (78)$$

The classical-shift subgroup acts trivially on the oscillator output; it only changes the classical outcome by $\sqrt{\gamma}s$. Setting $w = 0$, $x = 0$, and $s = x$, we obtain

$$\Lambda_E^x(X) = \Lambda_{E-\sqrt{\gamma}x}^0(X).$$

Define the residual instrument by $\mathcal{I}_E(X) := \Lambda_E^0(X)$. Then for a point input x , the output classical variable has the form $\sqrt{\gamma}x + r$, with residual instrument branch $\mathcal{I}(dr)$. Integrating the point-input identity against $x \sim \mathsf{N}_{0,V_p}$ gives (??), because $\sqrt{\gamma}X \sim \mathsf{N}_{0,\gamma V_p}$ and the residual branch is independent of the amplified Gaussian input.

Finally, setting $x = 0$ and using (??) with arbitrary w gives, for every Borel E ,

$$\mathcal{I}_E(D(w)XD(w)^*) = D(\sqrt{\gamma}w)\mathcal{I}_E(X)D(\sqrt{\gamma}w)^*.$$

Thus each Borel branch is displacement covariant with gain $\sqrt{\gamma}$. Trace preservation of Λ gives $\int \text{Tr}[\mathcal{I}(dr)(\Phi_0)] = \text{Tr} \Phi_0 = 1$. $\square$

Proposition A.2 (Hybrid converse from residual instruments). *The hybrid Gaussian product converse, ??, holds.*

Proof. By ?? for the amenable product group $\mathsf{H} \times \mathbb{C}^R$, it suffices to prove the origin bound for exactly product-covariant channels. Next, average such a channel over the per-mode phase group $\mathsf{U}(1)^R$: conjugate the oscillator input by $\bigotimes_{\alpha} R(\theta_{\alpha})$, conjugate the oscillator output by the same rotations, and integrate over θ with the Haar measure. Phase rotations act trivially on the classical registers and satisfy $R(\theta)D(w)R(\theta)^* = D(e^{i\theta}w)$, so this average preserves the exact $\mathsf{H} \times \mathbb{C}^R$ covariance with gain $\sqrt{\gamma}$; and since the origin input and the target below are phase invariant, joint concavity of root fidelity shows that the average does not decrease the origin fidelity. The averaged channel is phase covariant in every oscillator mode, and because the phase group does not move the classical outcome, the average is implemented branchwise on the associated instrument. Hence the residual branches $\mathcal{I}(dr)$ produced by ?? are displacement covariant with gain $\sqrt{\gamma}$ and phase covariant in every mode, which are exactly the hypotheses of ??; in particular the oscillator output at the origin is diagonal in the joint number basis.

Let

$$\Theta = \mathsf{N}_{0,V_p} \otimes \bigotimes_{\alpha} \tau_{q_{\alpha}}, \quad \Omega_* = \mathsf{N}_{0,\gamma V_p} \otimes \bigotimes_{\alpha} \tau_{q_{\alpha}^{(\gamma)}}$$

be the target and least-noise product output at the origin. In the variational formula of ??, the optimizer for the commuting pair (Ω_*, Θ) factors as $A_*(y, \mathbf{k}) = a_0(y)W_{\mathbf{k}}$, where

$$a_0(y) = \sqrt{\frac{d\mathsf{N}_{0,V_p}}{d\mathsf{N}_{0,\gamma V_p}}}(y), \quad W_{\mathbf{k}} = C \prod_{\alpha} \left(\frac{q_{\alpha}}{q_{\alpha}^{(\gamma)}} \right)^{k_{\alpha}/2}.$$

The function a_0 is a centered Gaussian decreasing in $\|y\|_{V_p^{-1}}$, and W is a bounded product of decreasing photon-number weights.

Let Ξ be the output of an exactly covariant hybrid channel at the origin. By ??,

$$\Xi(dy) = \int \mathbf{N}_{r,\gamma V_p}(dy) \mathcal{I}(dr)(\Phi_0).$$

For each residual r , the classical factor obeys

$$\int a_0(y) \mathbf{N}_{r,\gamma V_p}(dy) \leq \int a_0(y) \mathbf{N}_{0,\gamma V_p}(dy) = B_{\text{cl}}(\gamma, d),$$

since the left-hand side is a centered Gaussian function of r , maximized at $r = 0$. For the oscillator branch, ?? gives

$$\text{Tr}[\mathcal{I}(dr)(\Phi_0)W(N)] \leq \text{Tr}[\mathcal{I}(dr)(\Phi_0)] B_{\text{orb}}(\gamma, p).$$

Multiplying these two bounds and integrating over r yields

$$\text{Tr}(\Xi A_*) \leq B_{\text{cl}}(\gamma, d) B_{\text{orb}}(\gamma, p).$$

The other variational term is fixed by construction,

$$\text{Tr}(\Theta A_*^{-1}) = B_{\text{cl}}(\gamma, d) B_{\text{orb}}(\gamma, p).$$

Therefore

$$B(\Xi, \Theta) \leq B_{\text{cl}}(\gamma, d) B_{\text{orb}}(\gamma, p),$$

and squaring gives $F_{\text{cl}} F_{\text{orb}}$. The Hunt–Stein reduction gives the same bound for arbitrary channels in the flat-prior and expanding-window minimax limits. $\square$

Remark A.3 (What the residual form removes). *The possible obstruction was a classical residual correlated with a trace-decreasing quantum branch, analogous to a heralded amplifier. ?? shows that exact product covariance forces the amplified input Gaussian to be independent of the residual branch, and ?? bounds each covariant oscillator branch. Hence correlations between classical residual noise and quantum amplifier noise cannot improve the product fidelity.*

[Internal reference note: Keyl–Werner supplies Young-frame spectrum concentration; Kahn–Guta supplies finite-dimensional LAN with classical Gaussian and displaced-thermal oscillator components; Guta–Matsumoto supplies mixed Gaussian cloning via amplification and stochastic ordering; Li–Theil–Harrow–Chuang supply the random-purification comparison and QPA context used in ?. This note replaces the former “References to technical inputs” appendix and can be deleted once the bibliography and citation placement are finalized.]

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